

Per%Sense Web Port — Manual Test Plan

300 side-by-side test cases vs the legacy DOS application (100 Amortization · 100 Mortgage · 100 Present Value)

Purpose. This document is a manual acceptance test plan for the Go/web port of the Per%Sense DOS financial application. Each case gives concrete inputs to type into the web app, the behavior to verify, and the engine code path it exercises. The **DOS application is the financial authority**: unless a case says otherwise, "matches DOS" means running the identical inputs in the legacy DOS program side-by-side and confirming agreement to the cent (money) or displayed precision (rates/APRs). Do not compare against the Windows version or its help examples — several documented cases exist where the Windows help differs from actual DOS behavior.

Setup. Run the PerSense-Web binary and open it in a browser; run the DOS Per%Sense in DOSBox (or equivalent) beside it. Before every case, Clear All in the web app, reset the DOS worksheet, and set the computational settings identically in both programs. Dates are typed MM/DD/YYYY. Each section's intro (first page of the section) defines its default settings and any input shorthand used by that section's cases.

Recording results. Mark P (pass) or F (fail) in the P/F column and use Notes for the observed value or delta. For any failure, record BOTH the web output and the DOS output for the same inputs — a divergence can only be adjudicated against DOS. Cases flagged **DELIBERATE DIVERGENCE** or **[deliberate difference]** document places where the port intentionally does not reproduce a DOS bug or limitation (with the project doc that records the decision); do not file those as defects, but do record the DOS output in Notes.

Coverage. Each section's 100 cases are grouped: forward calculations across bases, frequencies and methods; backward (field-presence) solves for each solvable unknown; advanced options and combinations; errors, refusals and boundary values; and UI/workflow behaviors (green solved cells, hardening, undo, auto-calculate, CSV export). The Covers column names the Go engine function or branch each case exercises, so a failure can be routed straight to the responsible code.

Suggested order. Run each group's smoke cases first (the first few cases of each group are the simplest), then proceed to combinations. Groups are independent — sections can be split across testers, but keep any case that references an earlier case (e.g. "the solved payment P from AMZ-044") with the same tester.

Section	Cases	ID range	Screen
Amortization	100	AMZ-001 ... AMZ-100	Amortization (schedule, solvers, advanced options, payoff)
Mortgage	100	MTG-001 ... MTG-100	Mortgage worksheet (12-row grid, APR compare, What-If)
Present Value	100	PV-001 ... PV-100	Present Value (lump/periodic grids, COLA, VR, actuarial)

Section 1 — Amortization (AMZ-001 ... AMZ-100)

Setup. Run PerSense-Web (single binary; open the Amortization screen) side-by-side with the legacy **DOS** Per%Sense (the financial authority — do NOT compare against the Windows version or its help examples). Before every case: click **Clear All** in the web app, reset the DOS worksheet, and set the Computational Settings identically in both programs. Unless a case says otherwise, settings are at defaults: Basis days per year **360**, 1st interest prepaid at settlement **YES**, Interest paid in **Arrears**, Interest on interest **Actuarial**, Balloon / prepayment includes regular pmt **NO**, Exact method **NO**, Rule of 78s **NO**, Auto-calculate off. **BASE loan** shorthand = Amt Borrowed 100000, Loan Date 01/01/2025, Rate % 8, 1st Pmt Date 02/01/2025, # Periods 360, Pmts/Yr 12, Pmt Amount blank, Points 0. "Match DOS" means the computed payment, every schedule row (#, Date, Payment, Interest, Principal, Balance, Int to Date), and the totals line agree with the DOS app to the cent (spot-check the first 3 rows, any February, a mid-term row, the last 2 rows, and the totals). Where a case touches a **documented deliberate divergence**, the Expected column says so — do not file those as bugs, but do record the DOS output. Dates are typed MM/DD/YYYY.

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
Group A — Forward schedules: bases, frequencies, first periods, methods						
AMZ-001	Baseline 30/360 monthly, payment computed	BASE loan, all settings default	Payment fills green in Pmt Amount; 360 rows, no row 0 (natural first period → no settlement stub); final Balance 0.00; payment, every row, and totals match DOS to the cent	SolvePaymentClosedForm + generateSimpleSchedule		
AMZ-002	Second smoke: 15-yr loan	Amt 250000, Loan 06/01/2025, Rate 6.5, 1st Pmt 07/01/2025, # 180, Pmts/Yr 12	180 rows, balance 0.00, match DOS to the cent; Int to Date column is a running sum of interest	generateSimpleSchedule accumulation		
AMZ-003	Quarterly payments	Amt 50000, Loan 01/15/2025, Rate 7, 1st Pmt 04/15/2025, # 40, Pmts/Yr 4	40 rows dated quarterly on the 15th; balance 0.00; match DOS	GrowthPerPeriod perYr=4, AddPeriod dating		
AMZ-004	Annual payments	Amt 20000, Loan 03/01/2024, Rate 9, 1st Pmt 03/01/2025, # 10, Pmts/Yr 1	10 annual rows, match DOS; last date 03/01/2034	perYr=1 annual branch		
AMZ-005	Semi-monthly on 365 basis	Amt 60000, Loan 01/01/2025, Rate 6, 1st Pmt 01/16/2025, # 48, Pmts/Yr 24; Basis 365	48 rows on the 1st/16th; per-row interest varies with actual day count (Feb half-months smaller than Jan); match DOS to the cent	F2 semimonthly day-count rows, DaysCloseEnough ±half-month		
AMZ-006	Biweekly on 365 basis	Amt 120000, Loan 01/03/2025, Rate 7.25, 1st Pmt 01/17/2025, # 130, Pmts/Yr 26; Basis 365	130 rows every 14 days; balance 0.00; match DOS	perYr=26 day-count frequency path		
AMZ-007	Weekly on 360 basis → auto-coerced to 365	Amt 50000, Loan 01/06/2025, Rate 9, 1st Pmt 01/13/2025, # 104, Pmts/Yr 52; Basis 360	Both programs silently switch to a 365 basis (DOS shows a notice, Amortize.pas:297-303); results computed on 365 and match DOS; verify a notice/caption in the web app	handlers.go basisCoerced weekly/biweekly 360→365		
AMZ-008	365 basis, monthly	BASE loan; Basis 365	Clean month boundaries count as whole periods (first-row interest = a full period, not 31/365); rows and payment match DOS to the cent	firstPeriodProrate month-based fraction on 365		
AMZ-009	365/360 basis (rate kicker)	BASE loan; Basis 365/360	Matches the DOS app to the cent — the app scales the typed rate by 365/360 at the cell layer and the port reproduces this (discrepancies \$28); payment is higher than the 360-basis run	amzKickerRate handler kicker		
AMZ-010	Exact method on 365	BASE loan but # 120; Basis 365; Exact YES	True actual-day interest: February rows accrue visibly less than 31-day months; payment solved by iteration; rows + payment match DOS to the cent	exactDaily + iterated payment solve (dosIteratePayment)		
AMZ-011	Exact method inert on 360	BASE loan but # 120; Basis 360; Exact YES	Identical output to the same run with Exact NO — Exact is a no-op on the 360 basis in both programs (discrepancies \$8)	exactDaily basis gate		
AMZ-012	Exact on 365/360	BASE loan but # 120; Basis 365/360; Exact YES	Actual-day accrual plus the app-layer rate kicker; matches the DOS app to the cent	exactDaily × kicker interaction		
AMZ-013	Odd SHORT first period (AM Example 1)	Amt 100000, Loan 02/12/2024, Rate 8, 1st Pmt 03/01/2024, # 360, Pmts/Yr 12	Payment is augmented for the 19-day first period and matches DOS exactly. WARNING: the Windows help's unadjusted figure will NOT match — DOS is the authority (discrepancies \$7); prepaid YES adds a row-0 settlement-interest stub at the loan date	oddFirstPeriod + solveFancyPayment refine; PrepaidInterest stub		
AMZ-014	Long first period + prepaid stub	Amt 100000, Loan 03/15/2025, Rate 8, 1st Pmt 07/01/2025, # 360, Pmts/Yr 12	Row 0 settlement-interest stub dated 03/15/2025 covering the odd gap; 360 regular rows after it; payment augmented; match DOS in both the stub and rows	firstPeriodProrate odd-day stub, prepaid settlement row		
AMZ-015	Odd first period with prepaid OFF	Same as AMZ-014 but 1st interest prepaid NO	No row-0 stub; the extra partial-period interest rolls into row 1 (large first-row interest, balance may tick up once); the solved payment is the SAME as AMZ-014 (DOS augments in both modes, \$7); no negative-amortization note for the one-period bump	firstIntPrepaid=false path; A-W6 single-bump suppression		

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AMZ-016	Natural first period, prepaid YES → no stub	BASE loan (settings default)	No row 0 appears — a natural one-period gap has zero prepaid interest; identical to AMZ-001	PrepaidInterest zero-gap		
AMZ-017	Points: APR + settlement charge	BASE loan; Points 2	APR % fills (readonly, > Rate); loan-date settlement row includes the 2%×amount points charge and it flows into total interest (DOS Amortize.pas:1476-1491); APR matches DOS EstimateAndRefineAPRwithPoints	applyPointsSettlement + ComputeAPRwithPoints		
AMZ-018	Points 0: baseline APR always shown	BASE loan; Points 0	Web shows APR = the rate as an effective annual yield even at 0 points; DOS leaves APR blank for an untouched 0 — a documented presentation-only difference; schedule itself matches DOS	always-compute-APR handler note (vs Amortize.pas:1420)		
AMZ-019	In-advance (annuity-due), plain loan	BASE loan but # 120; Interest paid in Advance	Payment is lower than the arrears run; schedule retires to 0.00. NOTE: the plain (non-exact) in-advance schedule is a documented bounded frontier (envInadvPay) — small payment/row deltas vs DOS are expected; record the delta, do not fail sub-percent differences	needPaymentRefine in-advance annuity-due		
AMZ-020	Exact × in-advance × 365	BASE loan but # 120; Basis 365; Exact YES; Advance	DOS's distinct shape: a row-0 settlement-interest row at the loan date, first amortizing payment shifted one period later, n-1 actual-day amortizing rows; payment + every row match DOS to the cent (frontier doc, Revision 12)	generateExactInAdvanceSchedule + repayExactTerminal		
AMZ-021	Rule of 78s on 360	Amt 15000, Loan 01/01/2025, Rate 12, 1st Pmt 02/01/2025, # 48, Pmts/Yr 12; Rule of 78s YES	Same payment as the R78-NO run; interest/principal split front-loaded (early rows more interest); Int to Date and totals match DOS to the cent	R78 sum-of-digits split, F7 accumulator rounding		
AMZ-022	Rule of 78s inert on 365 basis	Same as AMZ-021 but Basis 365	Identical to the plain 365 run — R78 is silently inert on any non-360 basis in BOTH programs (DOS routes non-360 through RepayFancyLoan)	R78 Basis360 gate		
AMZ-023	R78 × in-advance: R78 wins	Same as AMZ-021 plus Interest paid in Advance	R78 split renders; payment solved as annuity-due; settlement row kept; match DOS (audit A4)	A4 R78-precedence branch		
AMZ-024	USA Rule negative amortization	Amt 50000, Loan 01/01/2025, Rate 12, 1st Pmt 02/01/2025, # 60, Pmts/Yr 12, Pmt Amount 400 (typed); Interest on interest USA Rule	Payment below interest → balance grows; unpaid interest kept in a separate balance with NO interest-on-interest; a negative-amortization note appears; rows match DOS with USA set	usap branch in RepayLoan/ComputeNext; A-W6 note		
AMZ-025	Actuarial negative amortization (contrast)	Same as AMZ-024 but Interest on interest Actuarial	Shortfall compounds into principal → balance grows FASTER than AMZ-024 at every row; match DOS with Actuarial set	actuarial neg-am compounding		
AMZ-026	Hard payment, residual fold + warning	Amt 100000, Loan 01/01/2025, Rate 8, 1st Pmt 02/01/2025, # 60, Pmts/Yr 12, Pmt Amount 733.76 (typed)	The 30-yr payment on a 5-yr term leaves a big residual: the LAST row absorbs it (payment ■ regular) and a warning about an implied terminating balloon appears; totals match DOS payhard run	TackOnFinalBalloon fold + appendScheduleWarnings A-W9		
AMZ-027	1st Pmt Date blank → defaulted	Amt 100000, Loan 03/15/2025, Rate 8, 1st Pmt blank , # 360, Pmts/Yr 12	1st Pmt Date fills green with 04/15/2025 (Loan Date + 1 period); schedule identical to typing it explicitly; match DOS	FirstPass A-FP-defFirst		
AMZ-028	Term derived from Last Pmt Date	Amt 100000, Loan 03/15/2025, Rate 8, 1st Pmt 04/15/2025, # Periods blank , Last Pmt Date 03/15/2055, Pmts/Yr 12	# Periods fills green with 360; schedule identical to AMZ-027; match DOS	FirstPass A-FP-n via NumberOfInstallments (A9 snap)		
AMZ-029	# Periods AND Last Pmt Date conflict → N wins	As AMZ-028 but # 360 and Last Pmt Date 01/01/2040 (inconsistent)	No error: # Periods wins and Last Pmt Date is rewritten to 1st + 359 periods (03/15/2055), matching DOS Amortize.pas:220-226 (audit A7)	A7 N-wins over-determined rule		
AMZ-030	Very long term (50 years)	Amt 100000, Loan 01/01/2025, Rate 7, 1st Pmt 02/01/2025, # 600, Pmts/Yr 12	600 rows, balance 0.00, totals match DOS; no horizon error (well under the 10000-payment cap)	long-schedule walk, MaxSchedulePeriods headroom		
Group B — Backward solves: amount, rate, balloon, prepayment, term, payoff						
AMZ-031	Solve Amt Borrowed (blank amount)	Amt blank , Loan 01/01/2025, Rate 7, 1st Pmt 02/01/2025, # 180, Pmts/Yr 12, Pmt Amount 1500 (typed)	Amt Borrowed fills green with the solved principal; schedule built from it retires to 0.00; solved amount + rows match DOS	SolveLoanAmount closed form		
AMZ-032	Amount solve, odd first + prepaid	Amt blank, Loan 01/10/2025, Rate 8, 1st Pmt 03/01/2025, # 120, Pmts/Yr 12, Pmt 900	Solved amount accounts for the settlement-stub first-period shift; matches DOS (audit A6 fix)	SolveLoanAmount prepaidNaturalStartShift		
AMZ-033	Amount solve on 365 basis	Same as AMZ-031 but Basis 365	Solved amount uses the 365 prorate (audit F3); matches DOS	needScheduleRefine non-360 prorate		
AMZ-034	Amount solve with skip months	Amt blank, Loan 01/01/2025, Rate 8, 1st 02/01/2025, # 60, P/Yr 12, Pmt 1200; Skip Months 7 - 8	Solved amount reflects the skipped months (audit A5 — option-aware terminal); schedule with zero-payment Jul/Aug rows retires to 0; match DOS	A5 hasFancyOptions amount solve		

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AMZ-035	Amount solve, in-advance	Same as AMZ-031 but Interest paid in Advance	Amount solves against the annuity-due schedule; NOTE: plain in-advance is the bounded envInadvPay frontier — record any small delta vs DOS, don't fail sub-percent	SolveLoanAmount in-advance refine		
AMZ-036	Solve Rate (blank rate)	Amt 100000, Loan 01/01/2025, Rate blank , 1st 02/01/2025, # 360, P/Yr 12, Pmt 733.76	Rate % fills green ≈ 8; schedule from the solved rate retires to 0; solved rate matches DOS solvedrate	SolveRate Iterate		
AMZ-037	Negative-rate solve (over-funded loan)	Amt 10000, Loan 01/01/2025, Rate blank, 1st 02/01/2025, # 60, P/Yr 12, Pmt 150	Total payments (9000) < principal → a NEGATIVE rate solves and displays (no clamp — discrepancies \$10 fix); matches DOS	SolveRate negative root		
AMZ-038	Rate solve with a balloon	Amt 100000, Rate blank, Loan 01/01/2025, 1st 02/01/2025, # 60, P/Yr 12, Pmt 1200; Balloon: Date 01/01/2030, Amount 50000	Rate solves so payment+balloon retire the loan; schedule shows the balloon on the last date; matches DOS (discrepancies \$12 fix)	SolveRate + solveFancyRate schedule-oracle bisection		
AMZ-039	Rate solve, R78 + odd first	Amt 20000, Rate blank, Loan 01/10/2025, 1st 03/01/2025, # 36, P/Yr 12, Pmt 650; Rule of 78s YES	Rate solves with the odd-first refine applied (P4-1 fix); match DOS	P4-1 R78 odd-first refine in SolveRate		
AMZ-040	Rate solve on 365/360 (kicker display)	Amt 100000, Rate blank, Loan 01/01/2025, 1st 02/01/2025, # 360, P/Yr 12, Pmt 750; Basis 365/360	Solved rate displayed is the DOS-app cell value (un-kickered for display, \$28); matches the rate the DOS app shows for the same inputs	amzUnkickerRate echo		
AMZ-041	Blank payment + user balloon → balloon ignored advisory	BASE loan (# 360, Pmt blank); Balloon: Date 02/01/2030, Amount 20000	Payment solves WITHOUT the balloon (a solved payment amortizes alone) and an advisory says the balloon was ignored / enter a Payment Amount; DOS behaves the same way computationally (no balloon row); advisory text is port-only	appendResultAdvisories A-W11		
AMZ-042	Target balloon solve (mid-term, blank balloon amount)	BASE loan but Pmt Amount 733.76 typed ; Balloon: Date 02/01/2030, Amount blank	Balloon Amount fills green ≈ the remaining balance at that date; schedule ends at the 02/01/2030 balloon row with balance 0.00; matches DOS EstimateAndRefineBalloon	SolveBalloonAmount		
AMZ-043	Terminating balloon on interest-only loan	Amt 100000, Rate 8, Loan 01/01/2025, 1st 02/01/2025, # 60, P/Yr 12, Pmt 666.67 typed; Balloon: Date 01/01/2030, Amount blank	Interest-only payment → solved balloon = the full principal; balance flat all term then 0.00 after the balloon; match DOS	SolveBalloonAmount interest-only boundary		
AMZ-044	Balloon solves to ~zero → advisory	Run BASE with # 60 blank-payment first and note the solved payment P; re-enter with Pmt Amount P typed , Balloon: Date on the last payment date (01/01/2030), Amount blank	Solved balloon = 0 and an advisory says the payment already retires the loan	A-W4 near-zero solved balloon		
AMZ-045	Negative solved balloon → advisory	Same as AMZ-044 but Pmt Amount P+50	Balloon solves NEGATIVE (DOS-faithful negative root, no clamp) and an advisory flags the over-payment	A-W5 negative balloon root		
AMZ-046	Prepayment amount solve (Stop Date bound)	BASE loan but Pmt 600 typed (# 360); Prepay row: Start 02/01/2026, Stop 01/01/2035, Per/Yr 12, Amount blank	Extra-payment amount fills green: the series that retires the loan on schedule despite the low payment; schedule retires to 0; match DOS AO9	SolvePrepaymentAmount / solvePrepayAmountAdditive		
AMZ-047	Prepayment amount solve (# Pmts bound)	Same as AMZ-046 but Stop Date blank, # Pmts 24	Amount solves for a 24-payment series; Stop Date derives and displays; match DOS	prepayStopDate NN-derived bound		
AMZ-048	Prepayment duration solve (no bound)	BASE loan but Pmt 733.76 typed; Prepay row: Start 02/01/2026, Per/Yr 12, Amount 500, Stop and # Pmts blank	The engine solves how long the extras must run: Stop Date fills green, loan retires early with balance 0.00; match DOS	SolvePrepaymentDuration + fillDerivedPrepayStops		
AMZ-049	Derive-only mode (Amount and Rate both blank)	Amt blank , Rate blank , Loan 01/01/2025, 1st 02/01/2025, Last 01/01/2035, # blank, P/Yr 12	No schedule and no totals; # Periods fills green with 120. Then clear Last, set # 120 → Last fills green 01/01/2035	handleAmortizationDeriveOnly + FirstPass both directions		
AMZ-050	Blank payment at Rate exactly 0	Amt 10000, Loan 01/01/2025, Rate 0 , 1st 02/01/2025, # 12, P/Yr 12, Pmt blank	Go solves the zero-interest annuity (payment = principal/12, zero interest rows). DOS emits NOTHING at exactly rate 0 (solver seed blowup — deliberate non-reproduction, F10b); rate 0.0000000001 works in both. Do not file the DOS blank as a port bug	F10b Teeny-rate branch		
AMZ-051	Term-from-payment attempt (UI gate)	BASE loan but Pmt Amount 733.76 typed, # Periods and Last Pmt Date both blank	The web UI refuses client-side: "Please fill in: # Periods or Last Pmt Date". DOS solves the term for this plain loan (DetermineLastPaymentDate). Record DOS's n — the engine has the A6 term solve but the UI gates it; log as a known UI limitation if confirmed	getAmzInput missing-field gate vs solveNPeriodsFromPayment		
AMZ-052	Payoff: date → balance	Calculate BASE first; then Payoff as of 07/15/2028	Balance field fills with principal owed PLUS interest accrued since the last payment (does not credit that date's payment — can exceed early-loan principal at high rates); matches DOS's balance query to the cent	PayoffBalance (ComputeBalanceFromDate)		
AMZ-053	Payoff: balance → date (inverse)	Calculate BASE first; then Balance 50000 (leave Payoff date empty, type into Balance)	Payoff-as-of date fills with the first payment date the balance drops through 50000, engine-solved; matches DOS ComputeDateFromBalance to the DAY (arrears is exact)	ComputeDateFromBalanceDOS + targetBalance wiring		
AMZ-054	Payoff inverse, in-advance	BASE but # 120, Interest paid in Advance; calculate; Balance 50000	Date fills; NOTE: the in-advance base-date-shift walk is documented as possibly one period off vs DOS (discrepancies \$13 remaining item) — record the delta, fail only if > 1 period	in-advance branch of ComputeDateFromBalanceDOS		

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AMZ-055	Payoff on Rule-of-78 loan	AMZ-021 inputs (R78 YES); calculate; Payoff as of 07/15/2026	Balance reflects the R78 rebate method, not straight actuarial; matches DOS payoff with R78 set; (USA-rule payoff was also fixed to apply usap — 20c)	payoff.go R78/usap branches		
Group C — Advanced Options: balloons, prepayments, ARMs, moratorium, target, skip, combos						
AMZ-056	Balloon ADD mode (default)	BASE loan but # 360, Pmt 733.76 typed; Balloon: Date 02/01/2030, Amount 20000; Balloon-incl NO	The 02/01/2030 row's Payment = regular + 20000; principal drops by the extra; loan retires before payment 360 with a final short row; match DOS	generateFancySchedule balloon merge, plus_regular		
AMZ-057	Balloon REPLACE mode	Same as AMZ-056 but setting Balloon / prepayment includes regular pmt YES	The balloon row's Payment = 20000 total (replaces the regular payment that period); match DOS	REPLACE-mode balloon (20b terminal fold family)		
AMZ-058	Off-cycle balloon (mid-month)	Same as AMZ-056 but Balloon Date 02/14/2027	The balloon appears as its OWN dated row on 02/14/2027 with partial interest accrued to that date; the next regular row accrues only from 02/14; match DOS (RepayFancyLoan exact-date application)	off-cycle balloon draining, periodYearFraction partial span		
AMZ-059	\$0 balloon in REPLACE mode (skipped installment)	BASE but # 60, Pmt typed as solved-payment from AMZ-044; Balloon: Date 06/01/2025, Amount 0 ; Balloon-incl YES	DOS treats \$0+YES as a skipped installment (F9). The web UI's tooltip promises the same, but the JS drops a 0 amount (the row may instead solve as a target balloon). Verify which happens — if a solved balloon appears instead of a skipped payment, log the tooltip/engine mismatch as a defect	F9 zero-balloon-REPLACE vs getAmzInput amount != 0 drop		
AMZ-060	Two balloons, different dates	AMZ-056 inputs plus a second Balloon: Date 02/01/2032, Amount 10000	Both balloons applied on their dates in sorted order; match DOS. SAFETY: never enter two balloons on the SAME date in DOS — it infinite-loops/hangs (F10e); the port computes sanely but the DOS side is unrecoverable	SortBalloons + F10e hang warning		
AMZ-061	Monthly prepayment series (additive)	BASE but # 360, Pmt 733.76 typed; Prepay: Start 06/01/2025, Stop 05/01/2026, Per/Yr 12, Amount 200	12 rows show Payment = regular + 200; loan retires slightly early; match DOS	prepayment merge in ComputeNext order		
AMZ-062	Semi-annual prepayment series	Same but Prepay: Start 06/01/2025, Stop 06/01/2028, Per/Yr 2, Amount 1000	An extra 1000 lands every 6 months on the series' own grid; match DOS	prepayment frequency grid		
AMZ-063	Prepayment REPLACE mode	AMZ-061 inputs but Balloon / prepayment includes regular pmt YES	Each series month's Payment = 200 total (replaces the regular payment — the setting applies to prepayments too); balance amortizes accordingly; match DOS	PlusRegular=false prepayment path		
AMZ-064	Off-cycle prepayment series	AMZ-061 but Start 06/14/2025 (mid-month)	Extra payments appear as their own dated rows on the 14th with partial-period interest; regular rows unaffected in date; match DOS	off-cycle prepayment drain rows		
AMZ-065	Prepay row with BOTH Stop Date and # Pmts	AMZ-061 inputs plus # Pmts 6 on the same row (Stop 05/01/2026 kept)	Schedule identical to AMZ-061 — the engine honors Stop Date only and ignores # Pmts when both are given (documented in the field tooltip); verify no error	prepayStopDate precedence		
AMZ-066	ARM: rate-only adjustment (AO5)	BASE but # 120; Adjustment: Date 02/01/2027, New Rate 9, New Amount blank	From 02/01/2027 the rate rises and the payment is re-solved over the remaining term (two payment levels visible); retires to 0.00; match DOS Re_Amortize	AO5 rate-only re-amortize		
AMZ-067	ARM: fully-specified rate + payment	BASE but # 120, Pmt 1213.28 typed; Adjustment: Date 02/01/2027, New Rate 9, New Amount 1300	Rate and payment both switch on the date; residual folds into the final row if under/over-funded; match DOS	fully-specified adjustment walk		
AMZ-068	AO6: payment-only adjustment (implied rate)	BASE but # 120, Pmt 1213.28 typed; Adjustment: Date 02/01/2027, New Rate blank , New Amount 1300	The engine solves the implied rate from the new payment (AO6); schedule continues at that rate; match DOS (P4-N6/N7 convergence fixes)	AO6 solveAdjRate		
AMZ-069	AO7: date-only adjustment (re-amortize at current rate)	BASE but # 120; Adjustment: Date 02/01/2027, Rate and Amount both blank	Payment re-solves at the unchanged rate — with a solved payment the schedule equals the no-adjustment run; no error; match DOS (AO7 alone is clean; see AMZ-072 for the buggy combo)	AO7 re-amortize signal		
AMZ-070	Multi-step ARM (3 adjustments)	BASE but # 120; Adjustments: 02/01/2026 rate 9; 02/01/2027 rate 10; 02/01/2028 rate 11 (amounts blank)	Three payment steps, each re-amortized; dates must be distinct and in-range; retires to 0.00; match DOS	SortAdjustments + sequential Re_Amortize		
AMZ-071	ARM + Points → APR	AMZ-066 inputs plus Points 3	APR % populates for the fancy loan and matches DOS (\$14 fix — fancy loans previously returned APR 0); settlement row includes the points charge	applyAPR on AmortizeDOS path		
AMZ-072	AO7 + later balloon — DELIBERATE DIVERGENCE	Amt 100000, Loan 01/01/2025, Rate 6, 1st 02/01/2025, # 24, P/Yr 12; Adjustment: Date 07/01/2025 only (rate+amount blank); Balloon: Date 01/01/2026, Amount 20000	Do NOT expect a DOS match. DOS hits a print-path corruption bug: retires the loan early with roughly HALF the interest, a bogus final-row date, and a "re-computed at 0.0000%" message. The port amortizes correctly to term with totals equal to the same loan without the adjustment (frontier doc "Deliberate divergences")	dosPortCanHandle AO6/AO7+balloon routing		
AMZ-073	Moratorium (interest-only period)	BASE but # 120, Rate 9; Moratorium 02/01/2026	Rows 02/01/2025–01/01/2026 are interest-only (Principal 0, flat balance), then full amortizing rows; retires to 0.00; match DOS	generateFancySchedule moratorium boundary		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
AMZ-074	Moratorium + blank payment at rate EXACTLY 10%	Amt 100000, Loan 01/01/2025, Rate 10 , 1st 02/01/2025, # 120, P/Yr 12; Moratorium 02/01/2026	Go solves the payment and retires to 0.00. DOS produces NO table at exactly 10.0000% (isolated Newton failure — deliberate non-reproduction, \$20g); 9.99% or 10.5% solve in both. Verify the 10.5% variant matches	20g moratorium payment solve; prepaidMoratoriumEarlyExit		
AMZ-075	Target: principal minimum	BASE but # 120; Principal minimum 600	Early rows where the natural principal component < 600 are raised so principal drops ≥ 600; later rows settle to the regular payment; retires to 0.00; match DOS	target minimum-principal branch (ComputeNext)		
AMZ-076	Skip months	BASE but # 60; Skip Months 6-8	Jun/Jul/Aug rows show Payment 0.00, interest accrues and balance rises those months; match DOS	MonthSetFromString + skip zeroing		
AMZ-077	Skip + blank payment solve	Same as AMZ-076, Pmt Amount blank	Solved payment is HIGHER than the no-skip run (it must cover the skipped months); schedule retires to 0.00; match DOS (skip solve routed through solveFancyPayment)	skip-month blank-payment solve, fancyOverUnder		
AMZ-078	Target OVERRIDES skip (DOS-faithful pair)	BASE but # 120; Skip Months 6-8 AND Principal minimum 600	In the skipped months the target's minimum-principal floor WINS over skip zeroing — those rows still reduce principal ≥ 600 (DOS AMORTOP.pas:643); match DOS	target-overrides-skip precedence		
AMZ-079	Balloon ON 1st Pmt Date, in-advance — DELIBERATE DIVERGENCE	BASE but # 60; Interest paid in Advance; Balloon: Date 02/01/2025 (= 1st Pmt), Amount 10000	Do NOT expect a DOS match. DOS's dead firstd init inflates the solved payment as if the balloon existed but never applies or collects it. The port applies the balloon, collects it in totals, and retires to -0 with the correct (lower) interest (frontier doc). Arrears variant of the same inputs DOES match DOS	balloon-on-first in-advance deliberate divergence		
AMZ-080	In-advance × skip × moratorium (fancy shape)	Amt 100000, Loan 01/01/2025, Rate 8, 1st 02/01/2025, # 48, P/Yr 12; Advance; Skip 6-8; Moratorium 02/01/2026	DOS in-advance-fancy shape: settlement-interest row at the loan date, first amortizing payment shifted one period, ordinary opening-balance interest on the shifted walk; payment + EVERY row match DOS to the cent (Revision 13 / P4-N1b)	inAdvanceFancy block, moratoriumActive shifted key		
Group D — Errors, refusals, boundaries						
AMZ-081	Missing required fields	BASE but clear Pmts/Yr and # Periods (Last Pmt Date also blank)	Client-side refusal listing exactly what's missing: "Please fill in: Pmts/Yr, # Periods or Last Pmt Date..."; nothing sent, no schedule	getAmzInput missing-field collection		
AMZ-082	Solve-mode without a payment	BASE but Rate blank and Pmt Amount blank	Refusal naming the payment: "Payment (needed to solve for the blank Rate)"; no schedule	solveMode payment guard		
AMZ-083	Loan Date after 1st Pmt Date	BASE but Loan Date 03/01/2025, 1st Pmt 02/01/2025	Error stating a payment would be due before the loan is made; DOS rejects the same dates-out-of-order case (parity)	ValidateInputs V6-9		
AMZ-084	Balloon before 1st Pmt Date	BASE but # 60; Balloon: Date 01/15/2025, Amount 5000	Error "Balloon ... dated before the 1st Pmt Date"; DOS refuses too ("Balloon cannot precede first regular payment") — error parity, wording differs; also holds with Interest paid in Advance	ValidateInputs C-A-4		
AMZ-085	In-advance + Rate Adjustment refused	BASE but # 120; Advance; Adjustment: Date 02/01/2027, Rate 9	Refusal: Rate Adjustments cannot combine with in-advance interest; DOS rejects the same combination (AMORTOP.pas:1294-1298) — parity	ValidateInputs V6-10		
AMZ-086	Unreachable principal Target	BASE but # 120; Principal minimum 1000 (> 100000/120)	Refusal that the Target exceeds Amount ÷ repaying periods; DOS refuses too (C-A-9); also retest with Moratorium 02/01/2026 — the floor uses post-moratorium period count	ValidateInputs C-A-9 nrepay		
AMZ-087	Rate solve beyond ±200%	Amt 10000, Rate blank, Loan 01/01/2025, 1st 02/01/2025, # 120, P/Yr 12, Pmt 6000	Refusal: rate did not converge / implies a rate beyond ±200% (DOS's Iterate brake, audit A10); no partial schedule	A10 rate>±200% iteration brake		
AMZ-088	Schedule size cap	BASE but # Periods 12000	Refusal citing the 10000-payment maximum; DOS's behavior at this size differs (its own date horizon) — wording/parity NOT expected, only that both refuse to produce 12000 rows	MaxSchedulePeriods guard		
AMZ-089	Skip Months parse errors	BASE but # 60; first Skip Months 6-13, then 6-	Each rejected with a specific message (month number 13 out of range; range dash with no end); no schedule; correct string 6-8, 12 then works	MonthSetFromString error arms		
AMZ-090	All twelve months skipped	BASE but # 60, Pmt blank; Skip Months 1-12	Both programs refuse (DOS: "did not converge"; port: a clearer all-months-skipped refusal — F10a). Wording differs; verify no all-zero growing schedule is emitted	allMonthsSkipped refusal		
AMZ-091	Single-payment loan (# Periods 1) — deliberate	Amt 10000, Loan 01/01/2025, Rate 8, 1st Pmt 02/01/2025, # 1, P/Yr 12	The port computes a one-row schedule retiring the loan (1st = Last date allowed); DOS treats firstDate = lastDate as an error — documented deliberate relaxation (validate.go C-A-5 note). Record DOS's refusal; don't file the port's success as a bug	C-A-5 degenerate one-payment relaxation		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
AMZ-092	Unusually-high-rate advisory	BASE but Rate 25	Schedule computes and matches DOS numerically; the port additionally shows a warning that ~25% is unusually high / check percent entry (fires above 20%) — advisory is port-only presentation	appendScheduleWarnings unusuallyHighRate		
Group E — UI & workflow						
AMZ-093	Auto-calculate on field exit	Settings → Auto-calculate ON; calculate BASE; change Rate 8 → 7.5 and press Enter (or Tab away)	Recalc fires automatically on leaving the field; payment and schedule update; while a field is mid-edit or input is incomplete, no error appears (silent auto mode)	autocalc hooks, autoSilent error suppression		
AMZ-094	Green solved cells + clear-to-solve workflow	Calculate BASE (Payment green). Press Calculate again. Then TYPE Pmt Amount 800 and CLEAR Amt Borrowed; Calculate	Second Calculate is idempotent (green cells re-read as blank — identical output). After the edit, Amt Borrowed fills green with the solved principal — clearing a field makes it the solved unknown	cell-output bookkeeping in getAmzInput; SolveLoanAmount dispatch		
AMZ-095	Typing over a solved payment changes dispatch	Calculate BASE; type 700 over the green Pmt Amount; Calculate	The typed value becomes a HARD payment: the schedule no longer retires evenly, the final row folds the residual, totals shift; matches a DOS payhard run of the same inputs	hard-payment over-determined dispatch		
AMZ-096	Undo	Calculate BASE; change Rate to 9; press Ctrl/⌘+Z (or the Undo button)	Rate returns to 8; Undo button disables when nothing is left to undo; a recalc reproduces the original results	pushUndo/undoLast stack		
AMZ-097	Clear All + Advanced badge + leftover-options hazard	Set a Balloon (02/01/2030, 20000) and Skip 6-8 on BASE; note the red count badge on Advanced Options; Calculate; press Clear All; press T in Loan Date	Badge shows the count of options in effect and the panel auto-expands; Clear All wipes the grid AND every advanced row/moratorium/target/skip and the badge; T inserts today's date	adv-badge, expandAmzAdvancedWithEntries, clearAmortization, T-key		
AMZ-098	Settings strip and schedule caption	Open Settings, set Basis 365; calculate BASE; then set Basis back to 360 and recalculate	The settings strip above the grid highlights the non-default Basis and clicking it opens Settings; the caption under the schedule records the settings the schedule was generated under; the two runs differ and each matches DOS under the same basis	amz-settings-strip, amz-schedule-settings caption		
AMZ-099	Export CSV	Calculate BASE; click Export CSV	A CSV downloads with a header plus 360 data rows; columns mirror the on-screen table (#, Date, Payment, Interest, Principal, Balance, Int to Date) and values equal the displayed schedule	exportAmzCSV		
AMZ-100	Quarterly / Yearly summary views	Calculate BASE; switch View to Quarterly summary, then Yearly summary	Aggregated rows (~120 quarterly / 30 yearly); each group's Payment/Interest/Principal sums equal the per-payment rows and grand totals are unchanged. NOTE: summary aggregation is presentation-grade (V6-14 scoped-down) — compare TOTALS with DOS's yearly view but treat per-payment rows as authoritative for grouping differences	renderAmzSchedule view aggregation (V6-14)		

Section 2 — Mortgage (MTG-001 ... MTG-100)

Tester setup: Run PerSense-Web and the legacy DOS Per%Sense side-by-side. All cases run on the web **Mortgage** screen (a 12-row grid with columns **Price, Points, % Down, Cash Required, Amt Borrowed, Years, Loan Rate, Mo Tax+Ins, Monthly Total, Balloon Yrs, Balloon Amt, APR**). Enter rate-type fields (**Loan Rate, % Down, Points**) as percent numbers (type 7 for 7%). Leave a field **blank** to have it solved; computed cells render green ("cell-output"). Toolbar buttons: **Calculate Row, Calculate All, Compare APR, What-If Table, Export CSV, Undo, Clear Row, Clear All**. Double-click (or press **H**) hardens a green cell into a fixed input; **C/F10** calculates the selected row; **Shift+Delete** clears a cell; **Ctrl/■+Z** undoes. Unless a case says otherwise, "matches DOS" means: type the identical inputs into the DOS Mortgage screen and confirm every solved cell agrees to the cent (money) or to displayed precision (percent/APR). Where a case touches a *known deliberate difference*, the Expected column says so explicitly — do not log those as defects. Reset with Clear All between cases unless the case builds on the prior one. Keep the Settings "Auto-calculate" checkbox **off** except where a case turns it on.

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
Group A — Single-row field-presence dispatch						
MTG-001	Solve Monthly from Price + % Down	Row 1: Price 100,000; % Down 20; Years 30; Loan Rate 7. Calculate Row	Cash Required, Amt Borrowed, Monthly Total fill as green computed cells; Amt Borrowed is 80% of Price (structural); Monthly matches DOS to the cent; APR column fills	Calc solve-monthly from Price+%Down; computeCashPctAndFinanced pct branch		
MTG-002	Solve Monthly with points and tax (help Ex 1)	Price 200,000; Points 2; % Down 20; Years 20; Loan Rate 8; Mo Tax+Ins 200. Calculate Row	Cash Required includes the 2-point charge on the financed amount; Monthly Total includes the \$200 tax; both match DOS to the cent	Calc solve-monthly with points+tax; Cash = down + points-financed		
MTG-003	Fund by Cash Required instead of % Down	Price 350,000; Cash Required 70,000; Years 30; Loan Rate 6.5. Calculate Row	% Down and Amt Borrowed solve (green), % Down = 20 (structural, points 0); Monthly solves; all match DOS	computeCashPctAndFinanced cash branch → solve-monthly		
MTG-004	Fund by Amt Borrowed	Price 250,000; Amt Borrowed 200,000; Years 30; Loan Rate 6. Calculate Row	% Down solves to 20 (structural) and Cash Required to the down payment; Monthly solves; matches DOS	computeCashPctAndFinanced financed branch		
MTG-005	Partial solve: Price + % Down only	Price 180,000; % Down 10 (nothing else). Calculate Row	Only Cash Required and Amt Borrowed fill; Monthly Total and APR stay blank; no error shown; matches DOS partial behavior	Calc partial path (no Years/Rate → main calc skipped)		
MTG-006	Partial solve: Price + Cash only	Price 180,000; Cash Required 18,000. Calculate Row	% Down and Amt Borrowed fill; Monthly stays blank; no error; matches DOS	cash branch without main calc		
MTG-007	Partial solve: Price + Amt Borrowed only	Price 180,000; Amt Borrowed 162,000. Calculate Row	% Down and Cash Required fill; Monthly blank; no error; matches DOS	financed branch without main calc		
MTG-008	Solve Price from Monthly + % Down	Monthly Total 1,400; % Down 20; Years 30; Loan Rate 6.5 (Price blank). Calculate Row	Price solves (green), then Cash Required and Amt Borrowed fill from it; all match DOS to the cent	Calc solve-price via pct branch		
MTG-009	Affordability via Cash (journey preset)	Cash Required 56,000; Points 1.5; Years 30; Loan Rate 8.5; Mo Tax+Ins 200; Monthly Total 1,650. Calculate Row	Price solves via the cash formula (points reduce buying power); % Down and Amt Borrowed fill; matches DOS to the cent	Calc solve-price via cash branch with points+tax		
MTG-010	Price-solve refusal: financed-only funding	Amt Borrowed 160,000; Years 30; Loan Rate 7; Monthly Total 1,100. Calculate Row	Refuses with error mentioning "% Down or Cash Required"; no cells computed; APR column stays empty (see MTG-058 for the compare implication); DOS shows an equivalent refusal	Calc solve-price refusal (no pct/cash funding)		
MTG-011	Tax is purely additive to Monthly	Repeat MTG-001 in Row 1; Row 2 same plus Mo Tax+Ins 250. Calculate All	Row 2 Monthly Total is exactly 250.00 above Row 1's; both rows match DOS	Monthly = P&I + Tax additivity		
MTG-012	Tax netted out in price solve	As MTG-008 plus Mo Tax+Ins 300. Calculate Row	Solved Price is lower than MTG-008's (only Monthly-Tax buys house); matches DOS	(Monthly-Tax)·Summation in solve-price		
MTG-013	Zero down typed explicitly	Price 150,000; % Down 0; Years 30; Loan Rate 7. Calculate Row	Accepted: Amt Borrowed = Price (structural), Cash Required shows 0 (typed zero honored, not treated as blank); Monthly matches DOS	zero-down valid; typed-0 vs phantom-0 UI rule		
MTG-014	Comma/dollar formatting parses	Price typed as "\$200,000.00"; % Down 20; Years 30; Loan Rate 7. Calculate Row	Identical results to typing 200000 plain; no parse error	UI parseMoney; input normalization		
MTG-015	High-rate warning threshold	Price 100,000; % Down 20; Years 30; Loan Rate 20 → Calculate; then change Loan Rate to 20.5 → Calculate	At 20: no warning. At 20.5: soft advisory "unusually high — double-check ... entered in percent" appears AND the row still computes; DOS shows its equivalent high-rate advisory; numbers match DOS	unusuallyHighTrueRate warning (DOS DA_UnusuallyHighRate), fire-only-above-threshold		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
MTG-016	Over-determined: Price + Monthly both typed	Price 200,000; % Down 20; Years 30; Loan Rate 7; Monthly Total 1,500. Calculate Row	Refuses with error "...nothing left to solve..."; web appends a hint about Balloon Yrs / clearing a field — richer wording than DOS's terse message (refusal itself matches DOS; wording difference is expected)	Calc over-determined branch; explainMtgError hint		
MTG-017	Empty row Calculate	Select an untouched row; Calculate Row	Nothing happens: no request, no error, no cells change	calcMortgageRow empty-body guard		
MTG-018	Quiet under-determined: Years + Rate only	Years 30; Loan Rate 7 (nothing else). Calculate Row	No outputs, no error (funding missing so main calc is skipped); DOS likewise computes nothing	hasFunding gate		
MTG-019	No anchor: Amt Borrowed + Years + Rate	Amt Borrowed 160,000; Years 30; Loan Rate 7 (no Price, no Monthly). Calculate Row	Nothing computed and no error — Calc needs Price or Monthly as anchor; confirm DOS behaves identically	Calc requires Price-or-Monthly anchor		
MTG-020	Monthly + funding but Years blank	Monthly Total 1,400; % Down 20; Loan Rate 7 (Years blank). Calculate Row	Nothing solves, no error; matches DOS	Years gate on main calc		
MTG-021	Rate blank blocks Monthly but not funding split	Price 200,000; % Down 20; Years 30 (Loan Rate blank). Calculate Row	Cash Required and Amt Borrowed fill; Monthly Total and APR stay blank; no error; matches DOS	Rate gate; partial funding solve still runs		
MTG-022	APR ≈ Loan Rate with no points	Row from MTG-001 (Points blank). Calculate Row	APR column shows ~7.00 (equals Loan Rate when Points = 0); matches DOS's APR display	FullTermAPR with zero points		
MTG-023	APR above Loan Rate with points	Price 200,000; % Down 20; Points 2; Years 30; Loan Rate 7. Calculate Row	APR is strictly greater than 7.0000; value matches DOS APR to displayed precision	FullTermAPR with points; IterateToFindAPR		
MTG-024	Points 0 typed vs Points blank	Row 1: MTG-023 inputs but Points 0; Row 2: same with Points blank. Calculate All	Both rows identical in every solved cell and APR	pointer-blank = 0 equivalence at API boundary		
MTG-025	Decimal rate entry	Price 300,000; % Down 20; Years 30; Loan Rate 6.875. Calculate Row	Monthly and APR match DOS to the cent / displayed precision	rate parsing + LoanRateToTrueRate boundary conversion		
MTG-026	Fractional % Down	Price 240,000; % Down 12.5; Years 30; Loan Rate 6.75. Calculate Row	Amt Borrowed is 87.5% of Price (structural); Monthly matches DOS	pct branch with fractional percent		
MTG-027	Jumbo realistic loan	Price 1,250,000; % Down 25; Years 30; Loan Rate 7.125. Calculate Row	All solved cells match DOS to the cent	Calc precision at large principal		
MTG-028	Term monotonicity 15 vs 30	Row 1: Price 200,000; % Down 20; Years 15; Loan Rate 6.5. Row 2: same, Years 30. Calculate All	Row 1 Monthly is strictly higher than Row 2's; both match DOS	Summation term dependence		
MTG-029	% Down back-out with points from Cash	Price 200,000; Points 2; Cash Required 44,000; Years 30; Loan Rate 7. Calculate Row	Solved % Down is <i>below</i> 22% (points portion of cash deducted first); Amt Borrowed and Monthly match DOS	cash branch formula (Cash/Price – Points)/(1 – Points)		
MTG-030	Cash from Amt Borrowed with points	Price 200,000; Amt Borrowed 160,000; Points 1; Years 30; Loan Rate 7. Calculate Row	Cash Required = down payment + 1% of Amt Borrowed (structural); matches DOS	financed branch + points in Cash		
MTG-031	Calculate All over three rows	Rows 1–3: MTG-001, MTG-003, MTG-008 inputs respectively. Calculate All	Each row solves exactly as it did individually; APRs fill where computable; one error-free pass	calcAllMortgageRows loop		
MTG-032	Quiet no-op: Monthly + Cash only	Monthly Total 1,650; Cash Required 56,000 (nothing else). Calculate Row	Nothing solves and no error (Years/Rate missing); matches DOS	main-calc gates with cash funding		
MTG-033	Selected-row scope of Calculate Row	Row 1: MTG-001 inputs; Row 2: MTG-003 inputs. Click into Row 2; Calculate Row	Only Row 2 solves; Row 1 remains all-input, un-computed	per-row dispatch; mtgSelectedRow scope		
MTG-034	Percent-units semantics (0.06 typed)	Price 100,000; % Down 20; Years 30; Loan Rate 0.06. Calculate Row	Treated as 0.06% (nearly interest-free): Monthly barely above Amt Borrowed/360; no warning; identical DOS entry gives identical result — record that the UI, like DOS, reads rates in percent	rate/100 scaling at UI boundary		
MTG-035	Full pipeline: five outputs from Cash funding	Price 300,000; Cash Required 45,000; Years 20; Loan Rate 6.75. Calculate Row	% Down, Amt Borrowed, Monthly Total and APR all fill; every one matches DOS	cash branch + solve-monthly + APR chain		
Group B — Balloon and Points variants						
MTG-036	Known balloon lowers Monthly	Price 200,000; % Down 20; Years 30; Loan Rate 7; Balloon Yrs 5; Balloon Amt 50,000. Calculate Row	Monthly solves <i>lower</i> than the same row without the balloon (compare vs MTG-001-style row); matches DOS to the cent	Calc balloonval discounting (BalloonKnown)		
MTG-037	Solve Balloon Amt from low Monthly	Price 200,000; % Down 20; Years 30; Loan Rate 7; Monthly Total 900; Balloon Yrs 5 (Balloon Amt blank). Calculate Row	Balloon Amt solves positive (green); matches DOS to the cent	balloonCalc (BalloonUnk path)		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
MTG-038	Balloon solve nets out tax	As MTG-037 but Monthly Total 1,100 and Mo Tax+Ins 200. Calculate Row	Solved Balloon Amt equals MTG-037's (engine uses Monthly–Tax); matches DOS	(Monthly–Tax) in balloonCalc		
MTG-039	Price solve with known balloon	Monthly Total 900; % Down 20; Years 30; Loan Rate 7; Balloon Yrs 5; Balloon Amt 50,000 (Price blank). Calculate Row	Price solves higher than without the balloon (discounted balloon adds buying power); matches DOS	solve-price with balloonval, pct branch		
MTG-040	Price solve via Cash + points + balloon	Cash Required 40,000; Points 1.5; Years 30; Loan Rate 7.5; Monthly Total 1,200; Balloon Yrs 7; Balloon Amt 30,000. Calculate Row	Price solves via the cash branch (with the (1–Points) factor); % Down/Amt Borrowed fill; matches DOS	solve-price cash branch with balloon+points		
MTG-041	Balloon timing changes Monthly	Row 1: MTG-036 inputs; Row 2: same but Balloon Yrs 10. Calculate All	Both compute; the two Monthly values differ (later balloon is discounted more); each matches DOS	Exxp(–rate–When) discounting per balloon date		
MTG-042	Balloon Yrs without Monthly: quiet	Price 200,000; % Down 20; Years 30; Loan Rate 7; Balloon Yrs 5 (Monthly and Balloon Amt blank). Calculate Row	Cash/Amt Borrowed fill but Monthly does not solve and no error — an unknown balloon needs a Monthly to size it; confirm DOS also computes no payment here	BalloonUnk blocks solve-monthly branch		
MTG-043	Harden Monthly, solve ~zero balloon (M-W1)	Solve MTG-001 row; double-click Monthly Total (hardens, flash); enter Balloon Yrs 5; Calculate Row	Balloon Amt solves to ≈ 0 and advisory "Balloon Amt is essentially zero..." appears. Advisory text is a port addition — DOS shows no message ; the near-zero number itself must match DOS	M-W1 advisory; harden-then-balloon-solve flow		
MTG-044	Negative balloon advisory (M-W2)	Price 200,000; % Down 20; Years 30; Loan Rate 7; Monthly Total 2,000; Balloon Yrs 5. Calculate Row	Balloon Amt solves negative ; advisory "Balloon Amt came out negative..." appears (port-only text); DOS computes the same negative number with no advisory	M-W2 advisory; over-paying monthly		
MTG-045	Negative amortization note (M-W3)	Price 200,000; % Down 20; Years 30; Loan Rate 7; Monthly Total 700; Balloon Yrs 10. Calculate Row	Balloon Amt solves larger than Amt Borrowed (payment below interest); note advisory about negative amortization appears (port-only); number matches DOS	M-W3 note; sub-interest payment		
MTG-046	Terminal balloon at final year (M-W7)	Price 200,000; % Down 20; Years 30; Loan Rate 7; Monthly Total 1,000; Balloon Yrs 30. Calculate Row	Balloon Amt solves to the remaining principal at year 30; M-W7 note about a terminal balloon appears (port-only text); number matches DOS	M-W7 note; Balloon Yrs = Years terminal case		
MTG-047	Balloon Yrs beyond term	Same as MTG-046 but Balloon Yrs 35. Calculate Row	Computes without refusal; M-W7 note appears (port-only); solved value matches DOS side-by-side	When > Years handling; M-W7 gate (When ≥ Years)		
MTG-048	Points rank APRs	Rows 1–3: Price 200,000; % Down 20; Years 30; Loan Rate 7; Points 0 / 1 / 3 respectively. Calculate All	APR strictly increases down the rows; Monthly Total identical on all three; each APR matches DOS	points→APR effect; APR ordering		
MTG-049	Points hit Cash, never Monthly	Row 1: MTG-001 inputs; Row 2: same plus Points 2. Calculate All	Row 2 Cash Required is higher by 2% of Amt Borrowed; Monthly identical between rows; both match DOS	points in Cash formula only		
MTG-050	Balloon + points + APR together	Price 200,000; % Down 20; Points 2; Years 30; Loan Rate 7; Balloon Yrs 5; Balloon Amt 50,000. Calculate Row	Monthly and APR both compute; APR reflects the balloon and points; matches DOS APR to displayed precision	IterateToFindAPR with balloon (TerminalBalloon path)		
Group C — Compare APR, What-If row generation, sorting						
MTG-051	Crossover comparison (points vs rate)	Row 1: Price 200,000; % Down 20; Points 3; Years 30; Loan Rate 8.1. Row 2: same but Points 1; Loan Rate 8.5. Calculate All; select Row 1; Compare APR	Panel shows both full-term APRs (match DOS) and "APRs cross at ...% for duration N years, M months. If held longer..." — crossover APR, duration and which loan wins all match DOS. Note: plural "1 years"/"1 months" wording is deliberately DOS-faithful — not a bug	CompareAPRs crossover; ite rateToFindCrossoverAPR and Time		
MTG-052	Dominance: one loan always better	Row 1: Price 200,000; % Down 20; Points 2; Years 30; Loan Rate 8.5. Row 2: same with Points 1; Loan Rate 8.0. Calculate All; Compare APR	Summary "Mortgage B (Row 2) is always better." (no crossover shown); verdict matches DOS	CompareAPRs always-better short-circuit (OneMonthAPR + full-term test)		
MTG-053	Tie: identical mortgages compared	Rows 1 and 2 identical: Price 200,000; % Down 20; Points 1; Years 30; Loan Rate 8. Calculate All; Compare APR	APRs shown equal; no "always better" verdict; whatever tie outcome appears (typically a non-converged/no-crossover report) must match DOS side-by-side — DOS is authority for tie behavior; no spurious crossover duration	CompareAPRs tie-break (strict-inequality guards)		
MTG-054	Compare anchors to selected row	Rows 1–3 all calculated (any three valid loans, e.g. rates 7 / 7.5 / 8). Click Row 3; Compare APR	Row 3 is reported as "Mortgage A (Row 3)"; partner is the next other APR-bearing row; row numbers named in the panel	compareMtgAPR selection anchoring (DOS cursor semantics)		
MTG-055	Compare with fewer than two APRs	Only Row 1 calculated (MTG-001). Compare APR. Then Clear All and Compare APR again	Both times: note "APR Comparison needs two calculated mortgages..." — no crash, no partial panel	compareMtgAPR <2-rows guard		
MTG-056	Compare across different terms	Row 1: Price 200,000; % Down 20; Years 15; Loan Rate 6.5. Row 2: same, Years 30; Loan Rate 7. Calculate All; Compare APR	Comparison runs (DOS deliberately allows unequal terms); APRs and verdict match DOS	CompareAPRs no same-term restriction		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
MTG-057	Compare with a balloon row	Row 1: MTG-036 inputs (balloon, solved Monthly); Row 2: MTG-001 inputs + Points 2. Calculate All; Compare APR	Comparison runs; if a crossover is reported it may sit exactly on the balloon year (balloon-date fallback); verdict and numbers match DOS	tryBalloonDates crossover fallback		
MTG-058	Amt Borrowed + Monthly rows: compare gate	Row 1: Amt Borrowed 160,000; Monthly Total 1,100; Years 30; Loan Rate 7. Row 2: Amt Borrowed 160,000; Monthly Total 1,080; Years 30; Loan Rate 6.75. Calculate All; Compare APR	Calculate shows the price refusal per row and APR cells stay empty, so the web Compare button reports "needs two calculated mortgages". Exact DOS match NOT expected: DOS performs this APR comparison (docs/discrepancies.md §17 — the backend /api/mortgage/compare accepts these rows; the toolbar gate requires APR-bearing rows). Record the behavior and the DOS APRs for reference	§17 EnoughDataForAPR vs UI APR-cell gate		
MTG-059	What-If: vary Loan Rate	Solve MTG-001 in Row 1; select it; What-If Table → Column "Loan Rate", Increment 0.25, # of lines 6; Generate	6 new rows appear below Row 1 with Loan Rate 7.25 → 8.50 (increment applied in loan-rate percentage points); Monthly re-solved per row and APR filled; each row matches the DOS row-generation table	GenerateRows VaryRate (yield-space step); EnoughDataForRowGeneration (Monthly is base output)		
MTG-060	What-If: vary Years with domain drop	Base as MTG-059; What-If → Column "Years", Increment -10, # of lines 4; Generate	Only rows for Years 20 and 10 appear (steps to 0 and -10 are dropped, never shown as garbage rows); surviving rows match DOS for those terms	GenerateRows VaryYears; whatIfStepValid domain filter (port guard — DOS dialog has no such filter; only the dropped-row behavior differs)		
MTG-061	What-If: vary Price	Base as MTG-059; What-If → Column "Price", Increment 25,000, # of lines 5; Generate	5 rows with Price 125,000 → 225,000; Cash/Amt Borrowed/Monthly re-solved per row; matches DOS	GenerateRows VaryPrice re-Calc chain		
MTG-062	What-If: two variables (2-D grid)	Base as MTG-059; What-If → var 1 "Loan Rate", inc 0.5, lines 2; second variable "% Down", inc 5, lines 1; Generate	(2+1)×(1+1)-1 = 5 new rows covering the rate × %-down grid; all rows calculate and match DOS combination-by-combination	GenerateGrid / client-side 2-D expansion		
MTG-063	What-If dialog validation	Select solved Row 1; What-If → leave Column "None" (or blank Increment); Generate	Error "What-If: specify a column, increment, and line count."; grid untouched	runWhatIf input validation		
MTG-064	What-If from insufficient source row	Select an empty row; What-If → "Loan Rate", 0.25, 5; Generate	Refuses with "What-If source row (row N) is missing: Price, % Down (or Cash Required or Amt Borrowed), Years, Loan Rate"; no rows created	rowgen insufficient data (frontend precheck of EnoughDataForRowGeneration inputs)		
MTG-065	Column sort 3-state with empties pinned	Rows 1–3 solved with different Monthly values; rows 4+ empty. Click "Monthly Total" header; click again; click a third time	1st click: ascending (arrow ▲), empty rows stay at the bottom; 2nd: descending, empties still bottom; 3rd: original order restored; # column rennumbers each time. (Sorting is a web-UI feature — no DOS comparison)	mtg sort asc→desc→unsorted cycle; empty-pinning		
MTG-066	Sort by APR; selection follows row	Rows from MTG-048 (APR 3-way spread). Select the Points 3 row; click APR header	Rows reorder by APR; the previously selected row keeps its highlight in its new position; Compare APR afterwards still anchors to that same mortgage	sort on computed APR column; selection tracking via original-row id		
Group D — What-if edits and UI workflow						
MTG-067	Blank a field to re-solve a different unknown	Solve MTG-001; then clear % Down (Shift+Delete) and type Amt Borrowed 150,000; Calculate Row	% Down and Cash Required re-solve from the new Amt Borrowed; Monthly updates; all match DOS for the new inputs	re-dispatch after changing which field is blank		
MTG-068	Single-field what-if edit	Solve MTG-001; change Loan Rate 7 → 6.5; Calculate Row	Monthly decreases; new values match a fresh DOS run at 6.5; old APR was cleared on edit and refills	re-solve after input edit		
MTG-069	Harden Cash, re-solve % Down	Solve MTG-001 (Cash green). Double-click Cash Required (green flash → input). Clear % Down. Calculate Row	% Down re-solves from Price+Cash; the hardened Cash value is byte-identical after recalc (hardened cells are invariant); results match DOS	hardenMtgCell; hardened-skip guard in updateMtgRowUI; cash branch		
MTG-070	Type over a computed cell	Solve MTG-001; type 900 over the green Monthly Total; add Balloon Yrs 5; Calculate Row	On typing, Monthly loses green styling and APR clears immediately; Calculate then solves Balloon Amt (row not over-determined because balloon is the unknown); matches DOS	onMtgCellEdit input-promotion; output→input transition		
MTG-071	Edits invalidate stale APR	Solve MTG-001 (APR filled); change Years 30 → 20 but do NOT calculate	APR cell empties the moment Years is edited (stale result never shown against new inputs)	APR-clear on cell edit		
MTG-072	Rows are independent	Row 1: MTG-001; Row 2: MTG-003. Calculate Row on Row 2 only, then corrupt Row 2 (Years 0) and Calculate Row	Row 1 never changes through any of it; Row 2's error does not touch Row 1's values	per-row status isolation (mtgStatus)		
MTG-073	Clear Row scope	Rows 1–2 solved; select Row 1; Clear Row	Row 1 fully blank (including APR and green styling, hardened flags); Row 2 untouched; error box cleared	clearMortgageRow		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
MTG-074	Clear All with confirmation	Grid with data (e.g. after MTG-059's What-If expansion). Clear All → Cancel; Clear All → OK	Cancel keeps everything; OK wipes all rows AND shrinks the grid back to 12 rows; confirm prompt only appears when data exists. (Confirmation dialog is web-only; no DOS analogue)	clearAllMortgageRows confirm + grid reset		
MTG-075	Undo restores a cleared row	Solve MTG-001; Clear Row; press Undo (or Ctrl/⌘+Z)	The row returns exactly as it was, including green computed styling	undo stack (pushUndo on clear)		
MTG-076	Undo removes a What-If expansion	Run MTG-059; press Undo once	All generated rows disappear; the source row survives untouched	what-if undo snapshot		
MTG-077	Keyboard shortcuts	In a filled row: press C, then F10 (after an edit); press Enter inside a field; Shift+Delete on a cell; H on a green cell	C and F10 both calculate the selected row; Enter moves focus to the next field; Shift+Delete blanks the focused cell; H hardens the selected computed cell (flash)	keyboard bindings per help-hint bar		
MTG-078	Auto-calculate mode	Settings → check "Auto-calculate". Type MTG-001's inputs, pressing Enter after each. Then blank % Down	Row solves by itself after the last needed field is committed (no button press); while the row is incomplete/stale, NO error appears — instead the "Showing a previous result..." hint may show; behavior mirrors DOS auto-recalc	autoCalcEnabled / autoSilent flow; setAutoCalcHint		
MTG-079	Export CSV	Solve rows 1–3 (MTG-031 setup); Export CSV	A CSV downloads containing the grid's rows with the displayed values (inputs and computed cells); values agree with the on-screen grid	exportMtgCSV		
MTG-080	Error row highlighting and recovery	Trigger MTG-016's over-determined error; then blank Monthly Total and Calculate Row	Error message is prefixed "Row N:" and the row-number cell is flagged (error highlight); after the fix, highlight and message clear and Monthly solves	markMtgErrorRow; error-clear on successful recalc		
Group E — Errors and refusals						
MTG-081	Years zero	Price 200,000; % Down 20; Years 0; Loan Rate 7. Calculate Row	Refuses with error "Years must be a positive whole number..."; nothing computed; DOS refuses equivalently	Calc Years ≤ 0 guard		
MTG-082	Years negative	Same but Years –5	Same refusal as MTG-081	Years guard, negative branch		
MTG-083	Amt Borrowed exceeds Price	Price 100,000; Amt Borrowed 120,000; Years 30; Loan Rate 7. Calculate Row	Refuses with exactly "Amount borrowed cannot exceed price." and no computed cells — this matches DOS's refusal (docs/discrepancies.md §16; oracle-verified). Any computed negative % Down would be a regression	Financed > Price refusal (Mortgage.pas:179-184 parity)		
MTG-084	Cash within 0.5% of Price	Price 100,000; Cash Required 99,700; Years 30; Loan Rate 7. Calculate Row	Refuses with error mentioning % Down rounding to 100%; no outputs; DOS refuses the same shape	cash branch pct ≥ 0.995 guard		
MTG-085	Amt Borrowed vanishingly small	Price 100,000; Amt Borrowed 400; Years 30; Loan Rate 7. Calculate Row	Refuses with "Amt Borrowed is too small next to Price..."; no outputs	financed branch pct ≥ 0.995 guard		
MTG-086	Price zero	Price 0; % Down 20; Years 30; Loan Rate 7. Calculate Row	Refuses with "Price must be greater than zero."; no outputs	computeCashPctAndFinanced zero-price guard		
MTG-087	Balloon Amt without Balloon Yrs	Price 200,000; % Down 20; Years 30; Loan Rate 7; Balloon Amt 50,000 (Balloon Yrs blank). Calculate Row	Refuses with error naming Balloon Yrs ("...enter Balloon Yrs... or clear Balloon Amt..."); no outputs; DOS refuses too (web wording richer — refusal itself is the parity point)	HowMuch-without-When guard		
MTG-088	Absurd rate typo (600)	Price 100,000; % Down 20; Years 30; Loan Rate 600. Calculate Row	High-rate advisory fires but the row still computes (soft warning, DOS-matching policy); the (extreme) Monthly matches DOS given the same literal entry	soft-warning-not-refusal on absurd rate		
MTG-089	Non-numeric input	Type "abc" in Price; % Down 20; Years 30; Loan Rate 7. Calculate Row	The unparseable field is treated as blank: no crash, no garbage — row behaves like MTG-018/021 (funding split can't run without Price; nothing solves, no JS error in console)	UI parse-failure → field omitted from request		
MTG-090	What-If runaway row count	Solved base in Row 1; What-If → "Loan Rate", 0.25, # of lines 500; Generate	Refuses with message that the grid is "capped at 200" total rows; no rows written; Undo not needed. (Port-only guard — the DOS dialog has its own screen limit; no exact-DOS comparison)	MTG_MAX_ROWS cap; MaxWhatIfRows backend guard		
Group F — Boundary values						
MTG-091	Tiny price	Price 1; % Down 20; Years 30; Loan Rate 7. Calculate Row	Computes without error; penny-level Cash/Amt Borrowed/Monthly match DOS exactly (rounding parity at the small end)	float64 parity at minimal magnitude		
MTG-092	Huge price	Price 100,000,000; % Down 20; Years 30; Loan Rate 7. Calculate Row	Computes; Monthly matches DOS to the cent (no float drift at 9 digits); APR matches	precision at large magnitude		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
MTG-093	Rate near zero	Price 120,000; % Down 0; Years 10; Loan Rate 0.0001. Calculate Row	Solves normally (no zero-rate error); Monthly a hair above the interest-free value; matches DOS (Teeny-cutoff not tripped)	Summation near the Teeny=1e-10 threshold		
MTG-094	Rate exactly zero	Price 120,000; % Down 0; Years 10; Loan Rate 0; Mo Tax+Ins 100. Calculate Row	Monthly = Amt Borrowed/120 + 100 exactly (zero-rate Summation branch = 12·t); matches DOS; APR column stays blank (rate fails the APR data gate) — record DOS's APR-cell behavior side-by-side	Summation r<teeny branch; EnoughDataForAPR OK(rate) gate		
MTG-095	Term of 1 year	Price 60,000; % Down 10; Years 1; Loan Rate 7. Calculate Row	12-payment loan solves; Monthly matches DOS to the cent; APR matches	minimal-term Summation		
MTG-096	Term of 50 years	Price 200,000; % Down 20; Years 50; Loan Rate 7.5. Calculate Row	Solves; Monthly approaches the interest-only floor but stays above it; matches DOS	long-term Summation / discount tail		
MTG-097	100% down	Price 250,000; % Down 100; Years 30; Loan Rate 7; Mo Tax+Ins 300. Calculate Row	Amt Borrowed solves to 0, Cash Required = Price, Monthly Total = 300 (tax only); no APR (no financed amount); confirm DOS does the same	pct = 1 degenerate funding; APR gate on Financed		
MTG-098	Cash just under / at the 99.5% cliff	Step 1: Price 100,000; Cash Required 99,400; Years 30; Loan Rate 7 → Calculate. Step 2: change Cash to 99,500 → Calculate	Step 1 computes (tiny Amt Borrowed ≈ 600, big % Down) matching DOS; Step 2 flips to the "would round to 100%" refusal — the boundary sits between the two	pct ≥ 0.995 threshold, both sides		
MTG-099	Balloon equal to Amt Borrowed	Price 200,000; % Down 20; Years 30; Loan Rate 7; Balloon Yrs 5; Balloon Amt 160,000. Calculate Row	Monthly solves sharply reduced but positive; no advisory (balloon was input, not solved); matches DOS to the cent	balloonval = full-financed discount edge		
MTG-100	Earliest balloon (year 1)	Price 200,000; % Down 20; Years 30; Loan Rate 7; Monthly Total 800; Balloon Yrs 1. Calculate Row	Balloon Amt solves large (little time to amortize); matches DOS to the cent; APR (with a filled row) matches DOS	balloonCalc at minimal When; Exxp(rate·1)		

Section 3 — Present Value (PV-001 ... PV-100)

Tester setup. Run PerSense-Web and the legacy DOS Per%Sense side by side and enter the identical scenario in both. Before every case: click **Clear All** on the web Present Value screen (the grids start with 4 blank rows; leftover rows are summed in silently) and reset **Settings** to defaults — Basis **360**, Exact method **NO**, COLA escalation month **Anniversary**, Interest paid in Arrears — unless the case says otherwise. Dates are MM/DD/YYYY. The three rate boxes (True Rate % / Loan Rate % / Yield %) are equivalent: editing one fills the other two — when comparing with DOS, read whichever rate convention the DOS screen is set to (the web **True Rate %** column matches the DOS/help "true rate" convention). To fire a backward solve, leave exactly one cell blank: a row-level target goes in that row's **Value** cell; a screen-level target goes in the **Present Value** box (with Rate or As-of Date blank). Solved cells render green after Calculate; refusals appear in the red error area under the grids. For actuarial cases select the **Per%Sense 1988** life tables — the SSA 2021 tables are new and are NOT expected to match DOS. "Matches DOS" means to the cent (or to DOS's displayed precision where DOS prints whole dollars). Any case marked [**deliberate difference**] touches a documented DOS-vs-port divergence (docs/discrepancies.md / dos_known_frontier.md) — an exact DOS match is NOT expected there; the Expected column says what to check instead.

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
Group A — Forward valuation, single- and multi-row						
PV-001	Basic lump-sum PV	Single Payments Row 1: Date 06/01/2026, Amount 50000. As-of Date 01/01/2024, Loan Rate % 6.0000. Present Value box blank	Row Value and Present Value total fill (green), equal to each other, and match DOS to the cent. Value < 50,000 (discounted)	forwardOnly + lumpRowPV, LineFullySpecified		
PV-002	Lump future value (as-of after date)	Row 1 (lump): Date 03/15/2024, Amount 25000. As-of Date 01/01/2027, Loan Rate % 6	Value > 25,000 (accumulation, negative YearsDif direction); matches DOS FV to the cent	forwardOnly accumulation direction		
PV-003	Lump on the as-of date	Row 1 (lump): Date 01/01/2024, Amount 10000. As-of Date 01/01/2024, Loan Rate % 6	Value = 10,000.00 exactly (zero discount period) in both apps	YearsDif=0 identity		
PV-004	Rate entered as True Rate	As in PV-001 but clear the rate boxes and type True Rate % 5.9850	Loan Rate % and Yield % auto-fill consistently (≈6.0000 / ≈6.1678); result matches PV-001 and DOS with its rate type set to True	\$4 rate interconversion, pvRateToTrue		
PV-005	Rate entered as Yield	As in PV-001 but type only Yield % 6.1678	True/Loan boxes auto-fill; total matches PV-004 to the cent (no lossy round-trip) and DOS in Yield convention	onPVRateEdit yield→true conversion		
PV-006	Negative lump (outflow)	Row 1 (lump): Date 06/01/2025, Amount -20000. Row 2: Date 06/01/2026, Amount 30000. As-of 01/01/2024, Loan Rate 6	Row 1 Value is negative, Row 2 positive; total = algebraic sum; matches DOS signs and cents	signed amounts through forwardOnly		
PV-007	Two lumps sum to total	Lump Row 1: Date 06/01/2025, Amount 15000; Row 2: Date 06/01/2026, Amount 15000. As-of 01/01/2024, Loan Rate 6	Present Value = Row1 Value + Row2 Value exactly; each row and the total match DOS	multi-row lump summation		
PV-008	Monthly annuity forward	Periodic Row 1: From Date 02/01/2024, Through 01/01/2027, Per/Yr 12, Amount 1000, COLA 0. As-of 01/01/2024, Loan Rate 6	36 payments valued; row Value = total, matches DOS to the cent	PeriodicSummation closed-form, SumFormula		
PV-009	Annual frequency	Periodic Row 1: From 06/01/2024, Through 06/01/2027, Per/Yr 1, Amount 12000. As-of 01/01/2024, Loan Rate 6	4 payments; matches DOS. Verify Per/Yr=1 payment count (both endpoint payments included)	NumberOfInstallments, peryr=1		
PV-010	Quarterly frequency	Periodic Row 1: From 03/01/2024, Through 12/01/2026, Per/Yr 4, Amount 2500. As-of 01/01/2024, Loan Rate 6	Matches DOS to the cent	peryr=4 realPerYr scaling		
PV-011	Biweekly frequency	Periodic Row 1: From 01/12/2024, Through 12/26/2025, Per/Yr 26, Amount 500. As-of 01/01/2024, Loan Rate 6	Matches DOS to the cent; payment count matches DOS	peryr=26 sub-monthly periods		
PV-012	Weekly frequency	Periodic Row 1: From 01/05/2024, Through 06/27/2025, Per/Yr 52, Amount 250. As-of 01/01/2024, Loan Rate 6	Matches DOS to the cent	peryr=52		
PV-013	Semi-annual frequency	Periodic Row 1: From 06/01/2024, Through 12/01/2026, Per/Yr 2, Amount 5000. As-of 01/01/2024, Loan Rate 6	Matches DOS to the cent	peryr=2		
PV-014	Future value of an annuity	Periodic Row 1: From 02/01/2024, Through 12/01/2025, Per/Yr 12, Amount 1000. As-of Date 06/01/2027, Loan Rate 6	As-of after the series: every payment accumulates; Value > 23×1000; matches DOS FV	PeriodicSummation with asOf > toDate		
PV-015	As-of inside the series	Periodic Row 1: From 02/01/2024, Through 01/01/2027, Per/Yr 12, Amount 1000. As-of Date 06/15/2025, Loan Rate 6	Payments before as-of accumulate, after discount; matches DOS (straddle -Inf branch)	straddling as-of / -Inf recompute branch		
PV-016	Mixed worksheet, 4 rows	Lump Rows 1-2: 06/01/2025 / 20000 and 09/01/2026 / 35000. Periodic Rows 1-2: (02/01/2024→01/01/2026, 12, 750) and (03/01/2024→03/01/2027, 4, 3000). As-of 01/01/2024, Loan Rate 6	Total = sum of the four green row Values; every row and total match DOS; pv-active-summary lists all four rows	multi-row forwardOnly aggregation		
PV-017	365 basis, lump	As PV-001 but Settings → Basis days per year = 365	Result differs from PV-001 and matches DOS with 365 DAY basis to the cent	Basis365 day-count in YearsDif		
PV-018	365 basis, periodic	As PV-008 but Basis 365	Matches DOS 365-basis to the cent	Basis365 periodic discounting		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
PV-019	365/360 kicker, lump	As PV-001 but Basis 365/360	Value is LOWER than the 360-basis PV-001 (more discounting). Matches the interactive DOS app to the cent. [deliberate difference] vs the DOS <i>headless oracle</i> only — the app applies the cell-layer rate kicker and the port follows the app (discrepancies \$27/\$28); side-by-side app comparison should match	pvKickerRate yield-space 365/360		
PV-020	365/360 periodic + COLA (COLA not kicked)	Periodic Row 1: From 02/01/2024, Through 01/01/2027, 12, 1000, COLA % 3. As-of 01/01/2024, Loan Rate 6, Basis 365/360	Matches DOS app to the cent; only the discount rate is kicker-scaled, the COLA is not (verify vs DOS with same COLA)	pvKickerRate excludes COLAcol		
PV-021	Exact method + 365 basis	As PV-018 (Basis 365) plus Settings → Exact method = YES	Result differs from PV-018 (period-by-period actual-day discount) and matches DOS with Exact=YES/365 to the cent (\$25)	PeriodicSummation exact loop		
PV-022	Exact on 360 basis is a no-op	As PV-008 (Basis 360) with Exact = YES	Total identical to PV-008 to the cent — Exact only takes effect with 365/365-360, matching DOS ("365 DAY MUST ALSO BE SELECTED")	Exact×Basis360 inertness		
PV-023	Exact + 365/360 combined	Lump: 10/01/2027, 500000. Periodic: 05/15/2025→12/15/2027, 12, 3500. As-of 01/01/2024, Loan Rate 5, Basis 365/360, Exact YES	Both rows and total match the DOS app to the cent (the \$27 repro shape). [deliberate difference] vs headless oracle only, as PV-019	pvKickerRate + exact loop together		
PV-024	Zero rate forward	Lump 06/01/2026 / 50000; Periodic 02/01/2024→01/01/2025, 12, 100. Rate boxes: enter 0 in Loan Rate %. As-of 01/01/2024	No discounting: lump Value = 50,000; periodic Value = 12×100; matches DOS with rate 0	rate=0 forward (below-Teeney discount inert)		
PV-025	30-year monthly series precision	Periodic: From 02/01/2024, Through 01/01/2054, 12, 1500. As-of 01/01/2024, Loan Rate 6	360 payments; total matches DOS to the cent (watch cumulative rounding)	long-n SumFormula stability		
Group B — Backward solves: six solver paths, dispatch, refusals						
PV-026	Solve lump Amount from row Value (PV-1)	Lump Row 1: Date 06/01/2026, Amount blank , Value 44000 (typed in the row's Value cell). As-of 01/01/2024, Loan Rate 6. Present Value box blank	Amount cell solves (green); solved Amount > 44,000; matches DOS. Row classified contains_unknown with Date present	solveLumpAmount, row-level target dispatch		
PV-027	Lump Amount from screen Sum Value residual	Lump Row 1: Date 06/01/2025, Amount 20000. Lump Row 2: Date 06/01/2026, Amount blank. As-of 01/01/2024, Loan Rate 6, Present Value box 60000	Row 2's Amount solves so the screen total hits 60,000; Row 1 stays as entered; matches DOS residual solve	computeKnownRowSum + solveLumpAmount via SumValue		
PV-028	Lump Amount solve, accumulation side	Lump Row 1: Date 03/01/2024, Amount blank, Value 30000. As-of 01/01/2027, Loan Rate 6	Solved Amount < 30,000 (payment earlier than as-of accrues up); matches DOS	solveLumpAmount negative-YearsDif direction		
PV-029	Solve lump Date (PV-2)	Lump Row 1: Date blank , Amount 50000, Value 44000. As-of 01/01/2024, Loan Rate 6	Date cell solves to a date after the as-of; matches the DOS-solved date (same day). Newton on date converges	solveLumpDate Newton		
PV-030	Lump Date solve lands before as-of	Lump Row 1: Date blank, Amount 40000, Value 50000. As-of 01/01/2026, Loan Rate 6	Value > Amount with positive rate ⇒ solved Date is BEFORE the as-of date; matches DOS	solveLumpDate sign/direction		
PV-031	Lump Date unreachable → refusal	Lump Row 1: Date blank, Amount 1000, Value 900000. As-of 01/01/2024, Loan Rate 6	Web refuses with a "date" computation ... did not converge style error (implied date falls outside 1900–2099); DOS reports non-convergence too. No nonsense date (e.g. 1596) may be displayed	solveLumpDate [1900,2099] window guard		
PV-032	Lump Date solve at zero rate → refusal	Lump Row 1: Date blank, Amount 50000, Value 44000. As-of 01/01/2024, Loan Rate % 0	Refusal that the rate is too small / at or near zero (with no interest, Amount and Value can never differ); DOS refuses likewise	solveLumpDate rate<Teeney guard		
PV-033	Solve periodic Amount (PV-4)	Periodic Row 1: From 02/01/2024, Through 01/01/2027, Per/Yr 12, Amount blank , Value 33000. As-of 01/01/2024, Loan Rate 6	Amount cell solves (Value ÷ summation factor); matches DOS to the cent	solvePeriodicAmount, LineContainsUnknown (From+To, no Amt)		
PV-034	Periodic Amount solve with COLA	As PV-033 plus COLA % 3	Solved Amount matches DOS (COLA-adjusted summation divisor)	solvePeriodicAmount + COLA factor		
PV-035	Periodic Amount residual on mixed screen	Lump Row 1: 06/01/2025 / 20000. Periodic Row 1: From 02/01/2024, Through 01/01/2026, 12, Amount blank. As-of 01/01/2024, Loan Rate 6, Present Value box 45000	The periodic Amount solves for the residual after the lump's value; the lump row is treated as known; matches DOS	SumValue residual + FirstPass fully_specified lump		
PV-036	Solve periodic Through date (PV-5)	Periodic Row 1: From 02/01/2024, Through blank , Per/Yr 12, Amount 1000, Value 33000. As-of 01/01/2024, Loan Rate 6	Through cell solves; matches the DOS date exactly (closed-form + ±1-period refinement)	solvePeriodicDate solveTo, Inn closed form		
PV-037	Through-date solve with COLA	As PV-036 plus COLA % 3, Value 36000	Solved Through matches DOS (Inn((last-f)/first))/(rate–cola) path with iteration)	solvePeriodicDate solveTo, cola≠0 branch		
PV-038	Through solve, COLA = rate equality branch	Loan Rate % 6.0000 (Yield shows ≈6.1678). Periodic Row 1: From 02/01/2024, Through blank, 12, Amount 1000, COLA % 6.1678, Value 24000. As-of 01/01/2024	Continuous COLA equals the true rate ⇒ every payment has equal PV; solver takes the AddNPeriods special case (n ≈ Value/per-payment-PV). Solved date matches DOS	solvePeriodicDate rate–cola<Teeney AddNPeriods branch		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
PV-039	Through solve unreachable target → refusal	Periodic Row 1: From 02/01/2024, Through blank, 12, Amount 100, Value 5000000. As-of 01/01/2024, Loan Rate 6	Target exceeds the perpetuity bound of a \$100/mo stream at 6% — web refuses (non-convergence / cannot-solve style error), DOS refuses too; no date past 2099 shown	solvePeriodicDate PUNT/non-converge path		
PV-040	Solve periodic From date (PV-6)	Periodic Row 1: From blank , Through 01/01/2027, Per/Yr 12, Amount 1000, Value 20000. As-of 01/01/2024, Loan Rate 6	From cell solves; matches DOS date (two-pass approximation)	solvePeriodicDate solveTo=false (BackwardPeriodicFrom)		
PV-041	From-date solve with COLA (second pass)	As PV-040 plus COLA % 3, Value 22000	Solved From matches DOS — the cola≠0 second refinement pass fires	PV-6 two-pass, cola≠0 second pass		
PV-042	Solve rate / IRR, single lump (PV-8)	Lump Row 1: Date 06/01/2026, Amount 50000. Clear ALL THREE rate boxes. As-of 01/01/2024. Present Value box: 44000	All three rate boxes solve and display consistently (True/Loan/Yield); solved rate matches DOS's solved rate in the same convention	solveRate Newton, BackwardRate dispatch		
PV-043	IRR on a periodic stream	Periodic Row 1: 02/01/2024→01/01/2027, 12, 1000. Rates blank, As-of 01/01/2024, Present Value 33000	Solved rate matches DOS; recomputing forward with the solved rate reproduces ≈33,000	solveRate periodic derivative path		
PV-044	IRR on mixed multi-row screen	Rows from PV-016 (2 lumps + 2 periodics), rates blank, As-of 01/01/2024, Present Value 70000	Rate solves against the whole-screen total; matches DOS	solveRate multi-row summation		
PV-045	Classic investment IRR (mixed signs)	Lump Row 1: 01/01/2024 / -100000. Periodic Row 1: 02/01/2024→01/01/2027, 12, +3200. Rates blank, As-of 01/01/2024, Present Value box 0	A sign-change cash flow with target 0: IRR solves; matches DOS. Verify the solved rate makes inflows and outflow cancel	solveRate sign-change root		
PV-046	Rate indeterminate → refusal	Lump Rows 1-2 entered as Date+Value only (06/01/2025 / Value 20000; 06/01/2026 / Value 30000). Rates blank, As-of 01/01/2024, Present Value 50000	Screen total is built from rate-independent pinned Values — web refuses with a "Rate is not determined ... specify amounts instead of values" style error, matching DOS's failure path	solveRate zero-derivative "not determined" refusal		
PV-047	Rate solve no-root → non-convergence	Lump Row 1: 06/01/2025 / 50000. Lump Row 2: 06/01/2026 / 50000. Rates blank, As-of 01/01/2024, Present Value box 0	All-positive flows can never total 0 — web errors "rate" computation did not converge (after the DOS-style second-pass restart from 0); DOS refuses likewise	solveRate 30-iteration restart + non-converge (PRESVALU.pas:736-745 parity)		
PV-048	Deep-discount IRR + high-rate advisory	Lump Row 1: Date 12/01/2026, Amount 100000. Rates blank, As-of 01/01/2024, Present Value 8000	Rate solves (very large, >100%); web additionally shows the P-W3 "solved rate is very high" note. Solved rate value itself matches DOS (DOS shows no advisory — advisory is web-only, cosmetic)	solveRate high-rate convergence + P-W3 advisory		
PV-049	IRR on 365/360 basis — display un-kick	As PV-043 but Basis 365/360, Present Value 32000	Solves; the DISPLAYED solved rate equals what the DOS app displays (the internal kicked rate is un-kicked for display). Re-running forward with the displayed rate reproduces the target. [deliberate difference] vs headless oracle only (§27)	pvUnkickerRate on solved-rate echo		
PV-050	Solve As-of Date, lump (PV-9)	Lump Row 1: Date 06/01/2026, Amount 50000. Loan Rate 6, As-of Date blank , Present Value box 44000	As-of Date solves (green) to a date before 06/01/2026; matches DOS's solved date	solveAsOf iterate, BackwardAsOf dispatch		
PV-051	Solve As-of Date, periodic	Periodic Row 1: 02/01/2024→01/01/2027, 12, 1000. Loan Rate 6, As-of blank, Present Value 30000	As-of solves; matches DOS date (asof += ln(target/sum)/rate iteration, ~2 passes)	solveAsOf log-step iteration		
PV-052	As-of solve at zero rate → refusal	As PV-050 but Loan Rate % 0	Refusal "cannot solve for the As-of Date because the Rate is too small (at or near zero)" style — DOS: "Cannot compute date - interest rate too small"	solveAsOf rate<Teeny refusal		
PV-053	As-of solve unreachable → non-convergence	Lump Row 1: Date 06/01/2026, Amount 50000. Loan Rate 6, As-of blank, Present Value box 50000000000	Implied as-of falls before 1900 — web errors "as of" computation did not converge; DOS refuses too. No pre-1900 date shown	solveAsOf maxdate/[1900,2099] guard		
PV-054	Multi-row dispatch: pinned row does not steal the solve (§26)	Lump Row 1: Date 06/01/2025, Value 20000 (Amount blank — Date+Value ⇒ known row, Amount derived). Periodic Row 1: From 02/01/2024, Through 01/01/2026, 12, Amount blank . As-of 01/01/2024, Loan Rate 6, Present Value 45000	The PERIODIC Amount is the solved unknown; the lump's Amount fills as a derived value from its pinned 20,000 Value (not from the residual). Matches DOS's choice of unknown	FirstPass Date+Value = LineFullySpecified (§26 fix), BackwardPeriodicAmount		
PV-055	Row with only a Value → refusal	Lump Row 1: Date blank, Amount blank, Value 25000 only. As-of 01/01/2024, Loan Rate 6	Refusal naming the row: only a Value is not enough — add Date (solves Amount) or Amount (solves Date). DOS similarly rejects. No solve is attempted	FirstPass LineContainsUnknown with only ValStatus, dispatch error path		
Group C — COLA						
PV-056	Monthly annuity + 3% annual COLA	Periodic Row 1: From 02/01/2024, Through 01/01/2027, Per/Yr 12, Amount 1000, COLA % 3. As-of 01/01/2024, Loan Rate 6, COLA month = Anniversary (default)	Value exceeds the no-COLA PV-008 result; matches DOS to the cent (stepped annual COLA, per-payment loop)	periodicSumAnnualCOLA anniversary stepping		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
PV-057	Annual frequency + COLA	Periodic Row 1: From 06/01/2024, Through 06/01/2027, Per/Yr 1, Amount 12000, COLA % 3. As-of 01/01/2024, Loan Rate 6	Matches DOS; with peryr=1 the closed-form continuous path applies (stepped mode requires peryr>1) — verify parity anyway	COLA yield→continuous Log1p, peryr=1 closed form		
PV-058	Continuous COLA setting	As PV-056 but Settings → COLA escalation month = Continuous	Result differs from PV-056 (proportional growth each payment); matches DOS with COLAmonth=CNT	COLAContinuous closed-form exp path		
PV-059	Month-specific COLA (January)	As PV-056 but COLA escalation month = January	COLA steps each January 1 rather than on the from-date anniversary; matches DOS COLAmonth=1	periodicSumAnnualCOLA month-specific coladate		
PV-060	COLA step lands once per year	Periodic Row 1: From 02/01/2024, Through 03/01/2025 (13 payments), 12, 1000, COLA 3. As-of 01/01/2024, Loan Rate 6. Then repeat with Through 01/01/2025 (12 payments)	Difference between the two runs reflects exactly one stepped payment at 1030 boundary behavior; both totals match DOS	colaAnniversary step timing (payment 13 carries the step)		
PV-061	Forward with COLA = rate	Loan Rate 6.0000; Periodic Row 1: 02/01/2024→01/01/2026, 12, 1000, COLA % 6.1678 (= displayed Yield). As-of 01/01/2024	No blow-up/NaN: growth and discount cancel; total matches DOS (near n × first-payment PV)	Inf=0 stability in SumFormula/stepped loop		
PV-062	Feb-29 From-date anniversary (leap)	Periodic Row 1: From 02/29/2024, Through 02/01/2027, 12, 1000, COLA 3. As-of 01/01/2024, Loan Rate 6	COLA anniversary stays anchored to the raw Feb-29 (steps when the Feb-29 boundary is reached, DOS un-normalized daterec — NOT clamped to Feb-28 or pushed to Mar-01). Total matches DOS to the cent (\$32 fix)	nextColaAnniversary raw inc(y), leap-day anchor		
PV-063	Month-end (Jan-31) anniversary	Periodic Row 1: From 01/31/2024, Through 01/01/2027, 12, 1000, COLA 3. As-of 01/01/2024, Loan Rate 6	Anniversary arithmetic uses the raw un-normalized date; step timing and total match DOS	raw-date anniversary vs AddYears clamping (\$29/\$32)		
PV-064	Perpetual stream with COLA ≥ rate → refusal	Periodic Row 1: From 02/01/2024, Through 12/31/2099 , 12, 100, COLA % 8. As-of 01/01/2024, Loan Rate 6	Both apps refuse: web error says the stream "runs forever has an infinite present value ... raise the Rate above the COLA"; DOS gives its rates≤COLA perpetual refusal. If instead both compute a finite number they must match — flag any asymmetry	infinite-PV guard, colaCont ≥ rate with toDate=latest		
PV-065	Finite band: rate between ln(1+cola) and cola (\$31)	As PV-064 but set True Rate % 7.9000 (COLA 8 ⇒ continuous COLA ≈ 7.696)	ACCEPTED (finite, very large PV) — the guard compares the continuous COLA. Must match DOS, which also accepts. A refusal here is a regression (\$31 pre-fix behavior)	perpetual guard continuous-COLA comparison band		
PV-066	Negative COLA (declining payments)	Periodic Row 1: 02/01/2024→01/01/2027, 12, 1000, COLA % -2. As-of 01/01/2024, Loan Rate 6	Value below the no-COLA PV-008 result; matches DOS with a negative COLA	Log1p(negative cola), stepped decline		
PV-067	COLA + Exact + 365	As PV-056 plus Basis 365, Exact YES	Matches DOS (exact per-payment loop with COLA multiplier) to the cent	exact loop × COLA (yrsFromStart-cola term)		
PV-068	COLA at biweekly frequency	Periodic Row 1: From 01/12/2024, Through 12/26/2026, Per/Yr 26, Amount 500, COLA 3. As-of 01/01/2024, Loan Rate 6	26/yr payments step once per anniversary year; matches DOS	stepped COLA × peryr=26		
PV-069	IRR with COLA rows	Periodic Row 1: 02/01/2024→01/01/2027, 12, 1000, COLA 3. Rates blank, As-of 01/01/2024, Present Value box 34000	Rate solves; matches DOS's solved rate; forward re-check reproduces target	solveRate with COLA-bearing summation		
PV-070	As-of solve with COLA row	Same row as PV-069, Loan Rate 6, As-of blank, Present Value 30000	As-of solves; matches DOS date	solveAsOf over COLA-adjusted sum		
Group D — Variable-rate schedules						
PV-071	Two-segment VR, lump	Variable Rate Schedule: Row 1 (starting) True Rate % 5; Row 2 Effective Date 01/01/2026, True Rate % 7. Lump Row 1: 06/01/2027, 100000. As-of 01/01/2024 (leave the top rate boxes at their defaults — they are ignored)	Discount integrates 5% to 01/01/2026 then 7% after; matches the DOS variable-rate (PVL) screen; settings strip/summary shows VR active	integrateRateForward + VRDiscountFactor 2-segment		
PV-072	Three-segment VR, periodic	Schedule: start 5; 01/01/2025 → 7; 01/01/2026 → 4. Periodic Row 1: 02/01/2024→01/01/2027, 12, 1000. As-of 01/01/2024	Per-payment discount crosses both boundaries; total matches DOS PVL to the cent	vrPeriodicValue multi-segment walk		
PV-073	VR entered via Loan Rate column	As PV-071 but type the schedule rates in the Loan Rate % column (5 and 7)	Converted client-side via 12·ln(1+r/12); matches DOS with the equivalent rate entries (compare in the same convention)	VR loan→true conversion precedence		
PV-074	VR boundary inside a payment period, straddling as-of	Schedule: start 6; 06/15/2025 → 8. Periodic Row 1: 02/01/2024→01/01/2027, 12, 1000. As-of Date 06/01/2025	Payments before as-of accumulate at the first rate, later ones discount across the mid-period change; matches DOS PVL	piecewise integration across as-of and mid-period boundary		
PV-075	Flat VR schedule ≡ fixed rate	Schedule: Row 1 True Rate % 5.9850 only (no other rows). Same worksheet as PV-008	Total equals the fixed-rate PV-008 run (Loan 6 ⇔ True 5.985) to the cent, and DOS PVL equals DOS PV likewise	VR single-segment identity		
PV-076	VR + stepped COLA	Schedule as PV-072; Periodic Row 1: 02/01/2024→01/01/2027, 12, 1000, COLA % 3. As-of 01/01/2024	Anniversary-stepped COLA multiplier composed with VR discount factors; matches DOS to the cent (raw-date anniversary per \$29/\$32)	vrPeriodicValue useStepped + colaAnniversary		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
PV-077	VR + life contingency	Schedule as PV-071. Periodic Row 1: 02/01/2024→01/01/2027, 12, 1000, Life = Living . Life Contingency: Person 1 Table = Per%Sense 1988 Male, Date of Birth 03/15/1958. As-of 01/01/2024	Each payment's VR discount is weighted by survival probability; matches DOS (VR+ACTU build) to the cent using the 1988 table	VR × LifeProb weighting		
PV-078	VR backward: solve Amount	Schedule as PV-071. Periodic Row 1: 02/01/2024→01/01/2027, 12, Amount blank . As-of 01/01/2024, Present Value box 33000	Amount solves by inverting the true VR valuation; matches DOS PVLX amount solve	solveVariableRateAmount (FancySummation inverse)		
PV-079	VR backward: solve lump Date	Schedule as PV-071. Lump Row 1: Date blank , Amount 100000. As-of 01/01/2024, Present Value box 85000	Date solves under the VR schedule; matches DOS PVLX date solve	solveVariableRateDate		
PV-080	VR mode: rate is never the target	Schedule as PV-071; Lump Row 1: 06/01/2027, 100000; clear all three top rate boxes; As-of 01/01/2024; Present Value box 85000 with NO blank row field	No IRR is attempted — the rate boxes are ignored in VR mode (help: "IRR cannot be computed in variable rate mode"); web computes forward / leaves target unsatisfied rather than solving a rate. DOS: "rates cannot be the target of a computation". Verify no bogus solved rate appears	VR guard: BackwardRate unreachable in RateSchedule mode		
Group E — Actuarial / life contingency						
PV-081	Single-life lump, Living	Lump Row 1: Date 06/01/2027, Amount 100000, Life = Living . Life Contingency: Person 1 Table = Per%Sense 1988 Male , Date of Birth 03/15/1958, Reference Date blank (uses as-of). As-of 01/01/2024, Loan Rate 6	Value = discounted amount × survival probability; matches DOS with its built-in table. (Selecting SSA 2021 instead is [deliberate difference] — a new table with no DOS counterpart; do NOT expect a DOS match there)	LifeProb Living, table parity ps88		
PV-082	Lifetime pension, periodic Living	Periodic Row 1: From 02/01/2024, Through 01/01/2044, 12, 2000, Life = Living. Person 1: 1988 Male, DOB 03/15/1958. As-of 01/01/2024, Loan Rate 6	Each payment weighted by survival to its date; total matches DOS to the cent	periodicWithActuarial per-installment weighting		
PV-083	Female table	As PV-082 with Table = Per%Sense 1988 Female	Higher value than PV-082 (longer survival); matches DOS female-table run	LifeProb sex-specific qx		
PV-084	Dead contingency	Lump Row 1: 06/01/2027, 100000, Life = Dead . Person 1: 1988 Male, DOB 03/15/1958. As-of 01/01/2024, Loan Rate 6	Value weighted by probability of death by the payment date; PV-081 value + PV-084 value = the uncontingent discounted value; matches DOS	Dead contingency complement		
PV-085	Age sensitivity	Run PV-082, then repeat with DOB 03/15/1944	Older DOB produces a strictly lower value; BOTH runs match DOS	LifeProb age conditioning		
PV-086	Two-life: Both Living	Periodic Row 1: 02/01/2024→01/01/2044, 12, 2000, Life = Both . Person 1: 1988 Male, DOB 03/15/1958; Person 2: 1988 Female, DOB 07/20/1960. As-of 01/01/2024, Loan Rate 6	Two-life options become selectable only once Person 2 is configured. Value weighted by joint survival (lowest of the two-life variants); matches DOS	two-life Both joint probability		
PV-087	Two-life: Either (survivor annuity)	As PV-086 with Life = Either	Value weighted by at-least-one-alive probability; strictly greater than PV-086's Both run; matches DOS	two-life Either union probability		
PV-088	Two-life: Only 1 / Only 2	Run PV-086's worksheet twice, Life = Only 1 , then Only 2	Each matches DOS; sanity: Only1 + Only2 + Both = Either row value (probability partition)	Only1/Only2 exclusive-survivor contingencies		
PV-089	Payment on Death (POD) value	No contingent rows needed. Life Contingency: Person 1 = 1988 Male, DOB 03/15/1958; Payment on Death (POD) 100000 . As-of 01/01/2024, Loan Rate 6. Optionally one plain lump row	The POD line under the panel shows the expected PV of the death benefit (integrated over the mortality distribution) and it is added into the total; matches DOS's POD value	PODValue mortality integration, POD-only forward path		
PV-090	Life + COLA pension	Periodic Row 1: 02/01/2024→01/01/2044, 12, 2000, COLA % 3, Life = Living. Person 1: 1988 Male, DOB 03/15/1958. As-of 01/01/2024, Loan Rate 6	Stepped COLA composed with survival weighting (raw-date anniversary — \$29/\$32 apply in the life path too); matches DOS to the cent	periodicWithActuarial × stepped COLA anniversary		
Group F — Errors and boundaries						
PV-091	Empty worksheet Calculate	Clear All; leave every grid row empty; As-of 01/01/2024 and Loan Rate 6 as defaults; press Calculate	Friendly refusal: "not enough inputs to solve this present value ... complete at least one payment row" style; DOS likewise refuses an empty screen. No crash, no 0.00 total presented as a result	not-Frontward/not-Backward error branch		
PV-092	Over-specified lump row	Lump Row 1: Date 06/01/2026, Amount 50000, AND type Value 40000. As-of 01/01/2024, Loan Rate 6	Soft warning ("row 1 is over-specified — the supplied Value is redundant and will be recomputed"), calculation proceeds, Value cell is REPLACED by the recomputed figure (= PV-001's), matching DOS's cancelable "value already determined by data above" behavior	FirstPass 3-field lump → warning + LineFullySpecified		
PV-093	Over-specified periodic row	Periodic Row 1: From 02/01/2024, Through 01/01/2027, 12, Amount 1000, AND Value 99999. As-of 01/01/2024, Loan Rate 6	Same pattern: warning, Value recomputed from dates+amount (= PV-008), total correct; DOS warns and proceeds	FirstPass periodic coreCount=3 + hasVal warning		
PV-094	Dates out of order	Periodic Row 1: From 01/01/2027, Through 02/01/2024, 12, 1000. As-of 01/01/2024, Loan Rate 6	Hard refusal naming the row: From Date must come before To Date (message also mentions the century setting); DOS refuses too	FirstPass DateComp order check		

ID	Title / Scenario	Inputs	Expected / Verify	Covers (code path)	P / F	Notes
PV-095	Zero Amount as the only row field	Lump Row 1: Amount 0 only (Date and Value blank). As-of 01/01/2024, Loan Rate 6, Present Value box 10000	Refusal: "the amount cannot be zero ... when it is the only field given"; likewise repeat with Value 0 only. DOS rejects both	C-P-2/C-P-3 zero-divisor guards		
PV-096	Date outside the 1900–2099 window	Lump Row 1: Date 06/01/ 1899 , Amount 50000 — then retry with 06/01/ 2150 . As-of 01/01/2024, Loan Rate 6	Both apps refuse or reject the date at entry (DOS daterec cannot represent years outside 1900–2099). Verify the web does NOT silently pivot the year to a different century or compute a value	date-window [1900,2099] validation parity		
PV-097	Rate at the Teeny cutoff	Lump Row 1: Date 06/01/2026, Amount 50000. As-of blank, Present Value 44000. Run with True Rate % 0.00000001 (= 1e-10 as a decimal, the Teeny cutoff), then 0.000001	At/below the cutoff: "Rate is too small" refusal (as PV-052). Just above it: either a solved date or a clean "did not converge" refusal — never a hang or an absurd date. DOS behaves the same at its teeny threshold	Teeny=1e-10 boundary in solveAsOf/solveLumpDate		
PV-098	Two-life contingency without Person 2	Attempt Life = "Only 2"/"Both" on a row with Person 2 Table = None	The four two-life options are DISABLED in the Life dropdown until Person 2 is configured; if a stale row retains one (configure Person 2, set Life=Both, then reset Person 2 to None), Calculate refuses naming the row and asking for the second life table/DOB — never a silent single-life collapse	refreshLifeContingencyOptions + checkSecondLifeProvided guard		
PV-099	55-year term	Periodic Row 1: From 02/01/2024, Through 01/01/2079, Per/Yr 12, Amount 1000. As-of 01/01/2024, Loan Rate 6	660 payments; matches DOS to the cent (no overflow/precision drift at long horizons)	long-horizon SumFormula / installment count		
PV-100	Extreme 65% rate	Lump Row 1: 06/01/2026, 50000; Periodic Row 1: 02/01/2024→01/01/2027, 12, 1000. As-of 01/01/2024, Loan Rate % 65	Computes without error; heavy discounting; both rows and total match DOS at the same 65% (verify rate-box interconversion still consistent at high rates)	high-rate Exxp/Lnn range, rate conversion at 50%+		